

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG

KHOA CÔNG NGHỆ THÔNG TIN I



BÁO CÁO
PHÂN TÍCH VÀ THIẾT KẾ HTTT

Tên đề tài: Restaurant Management System

Mã đề tài 20

Lớp: 05

Sinh viên: Lê Đức Mạnh - B22DCAT184

Subject No. 20

A restaurant management system enables management staff, sale staff and customers to use:

- Management staff: view statistics: dishes, ingredients, customers and suppliers. Manage dish information, make combo menus.
- Warehouse staff: import materials from suppliers, manage supplier information
- Sale staff: receive customers, take orders, receive payments at the table, make membership cards for customers, confirm table reservation information and order online.
- Customer: search, book a table and order food online.

Considering two modules and answering questions

- ***Customer searches for dish information:*** selects the menu to find dish information → enters the name of the dish to search → the system displays a list of dishes with the name of the keyword entered → clicks on a dish to view details → the system displays detailed information about the dish.
- ***Sale staff receives payment:*** selects the payment menu → finds the table by name → selects the correct table in the list of results → an invoice will appear containing the list of ordered dishes, unit price, total amount, total amount → confirms receive → prints invoices delivered to customers and receive money.

REQUIREMENT

A. Glossary list

No.	Name	Meaning
Human related		
1	Management staff	Management staff are the supervisors or managers of the restaurant. They use the system to monitor business performance, view statistics, and manage dishes and combo menus.
2	Warehouse staff	Warehouse staff are responsible for inventory control and supplier management. They use the system to import materials from suppliers and update supplier information.
3	Sale staff	Sale staff are employees who directly serve customers. They welcome guests, take orders, process payments, and print invoices. They also manage membership cards and confirm reservations, ensuring customers receive smooth and efficient service.
4	Customer	Customers are the end-users of the restaurant system. They can search for dish information, book tables, and order food online. Their role focuses on interacting with the restaurant's services in a convenient and user-friendly way.
5	Supplier	A supplier is an external party that provides ingredients, raw materials, or other goods to the restaurant. In the system, suppliers are managed by warehouse staff, who record supplier information and import materials from them.
Action related		
6	View statistics	See summarized information about dishes, ingredients, customers, and suppliers. This helps managers analyze business performance, monitor resources, and make informed decisions.
7	Manage dish information	Add, update, or remove details about dishes in the system. This includes editing the dish name, description, price so that the menu is always accurate and up to date for customers and sale staff.
8	Manage ingredients	Add, update, or remove information about ingredients used in the restaurant. This includes tracking ingredient names, quantities, and availability so that the kitchen always has the necessary materials for preparing dishes.
9	Manage customers	Handle customer records within the system. Sale staff can add new customers, update existing information such as names, contact details, and membership status, or remove

		inactive customers. This function supports customer service, loyalty programs, and efficient communication with clients.
10	Manage suppliers	Add, update, and maintain information about suppliers who provide ingredients and materials to the restaurant. This includes recording supplier names, contact details, and delivery history.
11	Make combo menus	Create special menu sets by combining multiple dishes into a package. This may include assigning a name, price, and description to the combo. The function helps promote sales, offer discounts, and provide customers with convenient meal options.
12	Import materials	Record the ingredients or supplies from suppliers into the system. This includes entering details such as material name, quantity, unit price, and supplier information.
13	Receive customers	Performed by sale staff to welcome guests when they arrive at the restaurant. This includes guiding them to available tables, confirming reservations if any, and ensuring they are comfortably seated.
14	Take orders	Record customers' food and drink requests into the system. This includes selecting dishes from the menu, noting special requirements, and sending the order to the kitchen.
15	Receive payments	Staff finalizes a customer's bill. They locate the correct table, review the invoice showing ordered dishes and total amount, then confirm and process the payment.
16	Make membership card	Create and issue membership cards for customers. This includes recording customer information such as name, contact details, and membership level.
17	Confirm table reservations	Verify and approve customer requests for booking tables. This includes checking table availability, updating the reservation status, and notifying customers of confirmation.
18	Confirm orders online	Staff validate customer food orders submitted through the online platform. They check the order details, ensure dish availability, and update the system with confirmation.
19	Search table online	Customers look for available tables in the restaurant through the online system. They can enter search criteria such as date, time, or number of guests, and the system displays a list of suitable tables.
20	Book table online	Customers reserve a table through the online system. After searching for available tables, the customer selects a preferred option, enters reservation details such as date, time, and number of people, then submits the booking request.
21	Order food online	Customers select dishes from the restaurant's online menu and place an order remotely. Customers can browse dishes,

		add them to a cart, and confirm their order through the system.
22	Search for dishes	Look up food items in the restaurant's menu by entering a keyword, such as the dish name. The system then displays a list of matching dishes.
23	Find dish information	Show details about dishes available in the restaurant. After entering a dish name or keyword, the system shows a list of matching results, and customers can select a dish to view its description and price.
24	Select menu	Action that involves choosing a menu category (such as appetizers, main dishes, or drinks) before performing a search.
25	Find table by name	Allows sale staff to access the payment section of the system. By choosing this option, they can manage customer bills, search for tables, review invoices, and process payments.
26	Print invoice	Allows sale staff to generate a physical copy of the customer's bill. The printed invoice includes details such as the ordered dishes, unit prices, total amount, and payment confirmation.
27	Receive money	Sale staff collect payment from customers after confirming the invoice. This may include cash, card, or other accepted payment methods.
28	Confirm receive payment	Validate and finalize a customer's payment in the system. Once the invoice is reviewed and the amount is collected, staff confirm the transaction so that it is officially recorded.
Object related		
29	Dish information	The details stored in the system about each menu item. This includes the dish name, description, ingredients, unit price, and availability status.
30	Menu	A collection of dishes offered by the restaurant, organized into categories such as appetizers, main courses, desserts, and drinks. In the system, the menu provides customers with options to browse, search, and select dishes, while staff use it to manage and update available items.
31	Table	A physical spot in the restaurant where customers are seated. In the system, each table is identified by a name or number and linked to customer orders and invoices.
32	Invoice	A document generated by the system that summarizes a customer's order and payment details. It includes information such as the list of ordered dishes, unit prices, quantities, total amount, and payment confirmation.

33	Ordered dish list	The section of an order or invoice that shows all the dishes a customer has requested. It includes the dish names, quantities, and corresponding unit prices.
34	Unit price	The cost assigned to a single serving or portion of a dish in the restaurant's menu. It is displayed in the ordered dish list and used to calculate the total amount on the invoice based on the quantity ordered.
35	Total amount	Final cost calculated for a customer's order. It is determined by summing the unit prices of all ordered dishes multiplied by their quantities, and may also include taxes or service charges. Total amount = $\Sigma(\text{Unit price} * \text{quantities}) + \text{tax} + \text{additional service}$
36	Keyword	A search term entered by the customer to find specific dishes in the restaurant's menu. When a keyword is provided, the system displays a list of dishes whose names or descriptions match the term.
37	Reservation	Information detailing a customer's request for a table. It includes customer name, contact info, date, time, number of guests, and status
38	Membership card	A physical or digital card issued to customers to track loyalty points or offer discounts. It links to customer information and their transaction history.

B. Project information

1. Purpose of the System

The Restaurant Management System is designed to support restaurant operations, improve customer service, and streamline staff tasks.

Its main purposes:

- Allow Management staff to control menus, suppliers, and statistics.
- Enable Warehouse staff to manage ingredients and suppliers.
- Assist Sale staff with customer service, order management, payment, and reservations.
- Provide Customers with online table booking, menu browsing, and ordering.

2. Scope of the System

Who can use the system and their permitted functions:

1. Staff

- Login
- Logout
- Change password

2. Management staff

- View statistics of dishes, ingredients, customers, suppliers.
- Manage dish information.
- Create combo menus.

3. Warehouse staff

- Import materials from suppliers.
- Manage supplier information.

4. Sale staff

- Receive customers.
- Take orders.
- Receive payments at the table.
- Issue membership cards.
- Confirm table reservations.
- Handle online orders.

5. Customers

- Search for dishes.
- Book tables online.
- Order food online.

3. Detailed Description of Two Modules

3.1. Customer searches for dish information

1. Customer opens the restaurant application or website.
2. Customer selects the “Search Menu” or “Dish Information” feature.
3. The system displays the searching interface.
4. Customer enters the name or keyword of the dish to search.
5. The system searches and displays a list of dishes matching the keyword.
6. Customer clicks on a dish from the search results to view details.
7. The system displays detailed information about the dish:
 - Dish name
 - Price
 - Ingredients
 - Image
 - Description
8. Customer reads the details and decides whether to search for another dish or exit the feature.

3.2. Sale staff receives payment

1. Sale Staff logs into the system.
2. Sale Staff selects the “Receive Payment” function.
3. The system displays the table search interface or a list of tables.
4. Sale Staff enters the table name or selects a table from the list.
5. The system displays the order information for that table:
 - List of ordered dishes
 - Unit price of each dish
 - Total quantity
 - Tax
 - Extra fee
 - Total amount
6. Sale Staff confirms the invoice with the customer.
7. Sale Staff clicks “Confirm Receive” in the system.
8. The system records the invoice to the database.
9. The system prints the invoice and delivers it to the customer.
10. Sale Staff receives the payment from the customer and completes the transaction.

4. Objects Managed in the System

4.1. Actor/User

Staff: username, password, full name, email, phone number

Management staff: inherit Staff, role

Sale staff: like management staff

Warehouse staff: like management staff

Customer: Name, email, phone number, membership level.

4.2. Entities

Module objects:

Menu: Name, description

Dish: Name, price, description, image

Table: Number, capacity, status

Order: Table number, date, dishes[], special note, status

Invoice: Order, total amount, date, status

Payment: Amount, payment method, date

Other objects:

Dish Statistics: Name, total sold, total revenue

Ingredient Statistics: Ingredient name, used quantity

Customer Statistics: Customer name, total orders, total spent

Supplier Statistics: Supplier name, total supplied, reliability score

Combo: Name, dishes[], price, description

Supplier: Name, phone number, email, address, reliability score

Membership Card: Customer name, points, tier

Online Order: Customer, dishes[], total amount, status

Table Reservation: Table, Customer, date, time, status

5. Relationships Between Objects

5.1. Actor/User Relationships

Staff (username, password, full name, email, phone)

- Management Staff inherits Staff (role = Management)
- Sale Staff inherits Staff (role = Sale)
- Warehouse Staff inherits Staff (role = Warehouse)

Customer (name, email, phone, membership level)

- Can have Membership Card (1:1 or optional)
- Can make Table Booking (1:N)
- Can place Online Order (1:N)

5.2. Module Object Relationships

Menu (name, description)

- Contains multiple Dish objects (1:N)

Dish (name, price, description, image)

- Can belong to Combo (M:N relationship, a dish can be in multiple combos)
- Can contained in Menu (M:N)
- One detail record can only contain one dish (Dish 1:N OrderDetail)

Table (number, capacity, status)

- Can have multiple Order objects (1:N)
- Can have multiple Table Booking objects (1:N)

Order (table number, date, dishes[], status, special note)

- Linked to Table (N:1)
- Contains multiple Dish objects (M:N, with quantity)
- Generates Invoice (1:1)
- One customer can have multiple order (Customer 1:N Order)

Invoice (order number, total amount, date, status)

- Associated with Payment (1:1 or 1:N if partial payments allowed)

Payment (invoice number, amount, payment method, date)

- Linked to Invoice (N:1)

5.3. Other System Objects / Statistics Relationships

Dish Statistics

- Associated with Dish (1:1)

Ingredient Statistics

- Associated with Ingredient (1:1)

Customer Statistics

- Associated with Customer (1:1)

Supplier Statistics

- Associated with Supplier (1:1)

Combo (name, dishes[], price, description)

- Contains multiple Dish objects (M:N)

Supplier (name, phone, email, address, reliability score)

- Supplies multiple Ingredients or Materials (1:N)

Membership Card (customer name, points, tier)

- Linked to Customer (1:1)

Online Order (customer, dishes[], total amount, status)

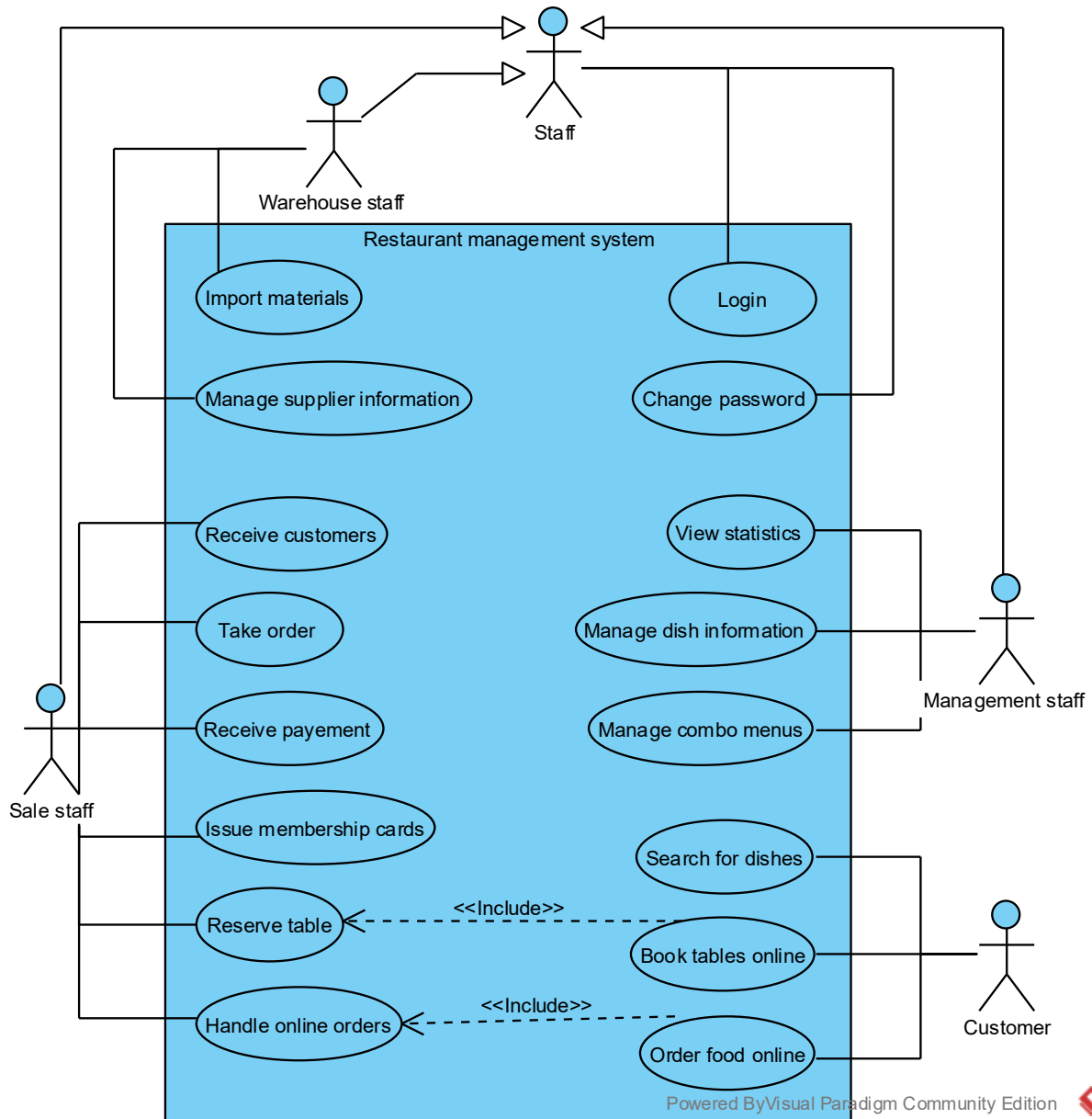
- Linked to Customer (N:1)
- Contains multiple Dish objects (M:N, with quantity)

Reservation (table, customer, date, time, status)

- Linked to Table (N:1)
- Linked to Customer (N:1)

C. Use case diagram

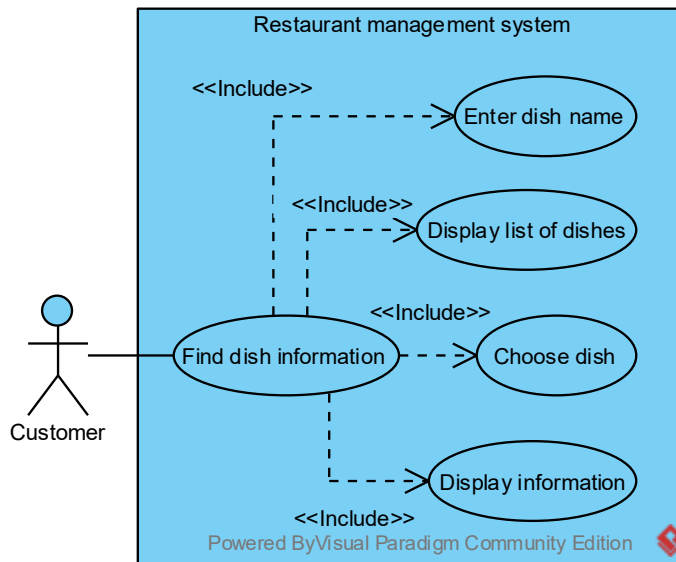
1. General use case diagram



- **Login:** Validates user credentials to ensure only authorized personnel can access the system
- **Change password:** Allows users to update their password when forgotten or outdated

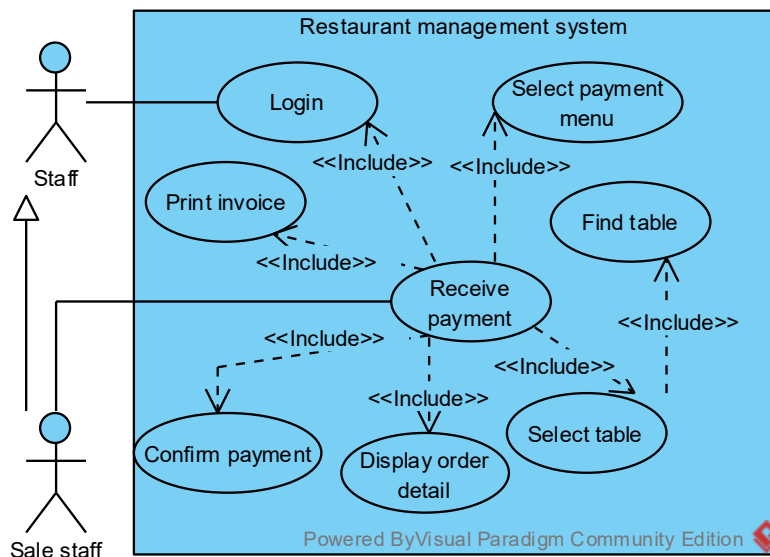
- **View statistics:** This UC allowss Management staff to see summarized information about dishes, ingredients, customers, and suppliers. This helps managers analyze business performance, monitor resources, and make informed decisions.
- **Manage dish information:** This UC allows Management staff to add, delete, edit the dish name, description, price so that the menu is always accurate and up to date for customers and sale staff.
- **Manage combo menus:** This UC allows Management staff to add, delete, edit combo meal in the menu
- **Import material:** This UC allows Warehouse staff to record the ingredients or supplies from suppliers into the system, entering details such as material name, quantity, unit price
- **Manage supplier information:** This UC allows Warehouse staff to add, update, and maintain information about suppliers who provide ingredients and materials to the restaurant
- **Receive customer:** This UC allows Sale staff to update tables status, confirming reservations if any
- **Take order:** This UC allows Sale staff to record customers' food and drink requests, note into the system
- **Receive payment:** This UC allows Sale staff to locate the correct table, review the invoice showing ordered dishes and total amount, then confirm and process the payment
- **Issue membership card:** This UC allows Sale staff to create and issue membership cards for loyal Customers
- **Reserve table:** This UC allows Customer to book a tabnle and Sale staff to verify and approve Customer online requests for booking tables
- **Handle online orders:** This UC allows Sale staff to validate Customer food orders submitted through the online platform and continue prossesing like site order
- **Search for dish:** This UC allows Customer to look up food items in the restaurant's menu by entering a keyword, such as the dish name
- **Book table online:** This UC allows Customers to reserve an available table through the online system
- **Order food online:** This UC allows Customers to select dishes from the restaurant's online menu and place an order remotely

2. Detail use case diagram for *Customer searches for dish information* module



- **Enter dish name:** This UC allows Customer to enter keyword to perform searches for dish information
- **Display list of dishes:** This UC allows System to reponse to Customer search using keyword
- **Choose dish:** This UC allows Customer to choose the dish they want from list of dish according to their search
- **Display information:** This UC allows System to display information about the dish Customer choose

3. Detail use case diagram for *Sale staff receives payment* module



- **Select payment menu:** This UC allows Sale staff to enter payment process
- **Select table:** This UC allows Sale staff to choose table for payment
- **Find table:** This UC help Sale staff to find desired table
- **Display invoice:** This UC allows Sale staff to to preview payment invoice
- **Confirm payment:** This UC allows Sale staff to confirm invoice with Customer
- **Print invoice:** This UC allows Sale staff to print physical voice and handle it to Customer

ANALYSIS

A. Scenarios

Use case	Searches for dish information			
Actor	Customer			
Preconditions	The customer has accessed the restaurant’s online system The system is running normally			
Postconditions	The customer views detailed information about the selected dish			
Main scenario	1. From the main interface, the customer selects the “Menu / Search dish” function			
	2. The system displays a interface with search box			
	3. The customer enters a keyword into the search bar (e.g., “Noodle”)			
	4. The system displays a list of dishes whose names match the keyword			
	No	Name	Price	Short description
	1	Beef Noodle	40.000 VND	Traditional beef noodle
	2	Chicken Noodle	40.000 VND	Noodle with chicken
	3	Hue Beef Noodle	45.000 VND	Hue origin beef noodle
	5. The customer clicks on one dish in the list			
	6. The system shows detailed information about the dish (name, price, full description, ingredients)			
	Name	Beef Noodle		
	Price	40.000 VND		
	Ingredients	Beef, salt, pepper, oil,bone broth, rice noodle		
	Description	...		
Exception	4. No dish matches the keyword 4.1 The system displays the message “No dish found”			

Use case	Receives payment																													
Actor	Sale staff																													
Preconditions	The system is running normally The sales staff has successfully logged into the system. There is a table requesting payment																													
Postconditions	The payment is recorded in the system The invoice is printed and the table status is updated to “Available”																													
Main scenario	1. From the main interface, the sales staff selects the “Payment” function.																													
	2. The system displays a interface with search box																													
	3. The staff enters the table number (e.g., “5”)																													
	4. The system shows a list of tables matching the search keyword																													
	<table><tr><th>Table</th><th>Status</th><th>Customer</th></tr><tr><td>5</td><td>Available</td><td>Null</td></tr><tr><td>15</td><td>Waiting for payment</td><td>Van An</td></tr><tr><td>25</td><td>In use</td><td>Nguyen Trung</td></tr><tr><td>35</td><td>In use</td><td>Le Nam</td></tr></table>			Table	Status	Customer	5	Available	Null	15	Waiting for payment	Van An	25	In use	Nguyen Trung	35	In use	Le Nam												
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	35	In use	Le Nam																											
5. The staff selects the correct table to process payment																														
6. The system displays the temporary invoice including ordered dishes, unit price, subtotal, tax (if any), and total amount																														
<table><tr><th>Dish name</th><th>Quantity</th><th>Unit price</th><th>Subtotal</th></tr><tr><td>Beef Noodle</td><td>2</td><td>40.000 VND</td><td>80.000 VND</td></tr><tr><td>Sping Rolls</td><td>1</td><td>30.000 VND</td><td>30.000 VND</td></tr><tr><td>Pepsi</td><td>1</td><td>10.000 VND</td><td>10.000 VND</td></tr><tr><td colspan="3">Dish total</td><td>120.000 VND</td></tr><tr><td colspan="3">VAT (10%)</td><td>12.000 VND</td></tr><tr><td colspan="3">Total</td><td>132.000 VND</td></tr></table>			Dish name	Quantity	Unit price	Subtotal	Beef Noodle	2	40.000 VND	80.000 VND	Sping Rolls	1	30.000 VND	30.000 VND	Pepsi	1	10.000 VND	10.000 VND	Dish total			120.000 VND	VAT (10%)			12.000 VND	Total			132.000 VND
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VAT (10%)			12.000 VND																											
Total			132.000 VND																											
7. The staff reviews the invoice and confirms the payment																														
8. The system records the transaction, prints the invoice, and updates the table status to “Available”																														
9. The staff gives the printed invoice to the customer and receives payment (cash/card)																														
Exception	4. No table found 4.1 The system shows the message “No matching table found” 5. If the invoice information is incorrect 5.1 The staff reports to the manager for correction before confirming payment 5.2 After confirmation, staff continue 6																													

B. Class diagram

1. System description

The Restaurant Management System is designated to support four key user roles: Management Staff, Warehouse Staff, Sales Staff, and Customers.

- **Management Staff:** Responsible for analyzing business performance. They can view statistics related to dishes, ingredients, customers, and suppliers. Additionally, they manage the menu by adding, updating, or removing dish information and creating combo menus.
- **Warehouse Staff:** In charge of inventory control. Their tasks include importing materials from suppliers, updating stock levels, and managing supplier records (contact details, supplied materials).
- **Sales Staff:** Direct point of contact for customers. They handle table reception, take orders, issue invoices, and process payments. Furthermore, they manage membership cards, confirm table reservations, and validate online orders.
- **Customers:** Interact with the system online. They can search for dishes by keyword, view detailed information (price, description, ingredients), book tables, and place online orders.

Key Scenarios:

- *Search:* A customer selects the menu function and enters a keyword. The system displays matching dishes; selecting one reveals its details.
- *Payment:* Sales staff selects the payment function, locates the specific table, and views the auto-generated invoice (items, unit prices, total). Upon confirmation, they print the invoice and process the payment.

2. Extract nouns

a. Actors: Staff (Abstract/Parent), Management Staff, Warehouse Staff, Sales Staff, Customer

b. Entities / Objects: Menu, Dish, Combo, Supplier, Table, Reservation, Order, Online Order, Invoice, Payment, Membership Card, Statistics

c. Attributes / Information

- **Dish related:** Name, Price, Description, Image.
- **Inventory related:** Unit, Import date.

- **Supplier related:** Name, Contact details (Email/Phone).
- **Invoice/Payment related:** Total amount, Unit price, Tax, Created date, Payment status.
- **Membership related:** Loyalty points, Tier.

3. Evaluate and choose nouns as entity classes or attributes

3.1. People (Actors)

- **Customer:** name, email, phoneNumber
- **Staff (Abstract):** username, password, fullName, email, phoneNumber
 - **ManagementStaff:** inherits Staff, position
 - **WarehouseStaff:** inherits Staff, position
 - **SalesStaff:** inherits Staff, position

3.2. Menu & Dishes Management

- **Menu:** name, description
- **Dish:** name, description, price, image
- **ComboMenu:** name, description, price

3.3. Warehouse Management

- **Supplier:** name, phoneNumber, email, address, reliabilityScore
- **SupplyEntry:** importDate, totalCost
- **SupplyDetail:** quantity, unitPrice

3.4. Sales & Order Management

- **Table:** number, capacity, status
- **Reservation:** date, time, numberOfGuests, status
- **MembershipCard:** issueDate, level, points
- **Order:** orderDate, status, specialNote
- **OnlineOrder:** inherits Order, deliveryAddress, contactPhone
- **OrderDetail:** quantity, unitPrice, subtotal
- **Invoice:** createdDate, totalAmount, status
- **Payment:** paymentDate, paymentMethod, amount

3.5. Statistics

- **DishStat:** period, totalSold, totalRevenue
- **CustomerStat:** period, totalReservations, totalOrders, totalSpent

4. Identify relationships

4.1. Customer - MembershipCard

- Relationship: 1 - 0..1
- Type: Association (a customer can optionally have one membership card)

4.2. Customer - Reservation

- Relationship: 1 - N
- Type: Association (a customer can make many reservations).

4.3. Customer - Order

- Relationship: 1 - N
- Type: Association (a customer can place multiple orders, including online orders).

4.4. Order - OnlineOrder

- Relationship: Inheritance
- Type: Generalization (OnlineOrder extends Order with delivery info).

4.5. Staff - ManagementStaff / WarehouseStaff / SalesStaff

- Relationship: Inheritance
- Type: Generalization (three staff types inherit from abstract class Staff).

4.6. Menu - Dish

- Relationship: 1 - N
- Type: Aggregation (a menu contains multiple dishes)

4.7. WarehouseStaff - Supplier / SupplyEntry

- Relationship: 1 - N (warehouse side)
- Type: Association (Warehouse staff records suppliers, and supply entries).

4.8. Dish - ComboMenu

- Relationship: N - N
- Type: Aggregation (many dishes can belong to many combo menus).

4.9. SalesStaff - Customer / Table / Reservation / Order / Invoice / Payment

- Relationship: 1 - N (sales side)

- Type: Association (Sales staff interacts with customers, manages reservations, takes orders, issues invoices, and processes payments).

4.10. Supplier - SupplyEntry

- Relationship: 1 - N
- Type: Association (a supplier can have many supply entries for deliveries).

4.11. Table - Reservation

- Relationship: 1 - N
- Type: Association (a table can be reserved multiple times).

4.12. Table - Order

- Relationship: 1 - N
- Type: Association (a table can have many orders during its usage).

4.13. Order - Dish (via OrderDetail)

- Relationship: N - N
- Type: Association (an order can contain multiple dishes, and each dish can appear in multiple orders).
- Implementation: requires intermediate entity OrderDetail.

4.14. Order - OrderDetail

- Relationship: 1 - N
- Type: Association (each order has multiple order details for items).

4.15. Order - Invoice

- Relationship: 1 - 1
- Type: Association (each order generates exactly one invoice).

4.16. Invoice - Payment

- Relationship: 1 - N
- Type: Aggregation (an invoice can have one or many payments, in case of partial payments).

4.17. Dish - DishesStat

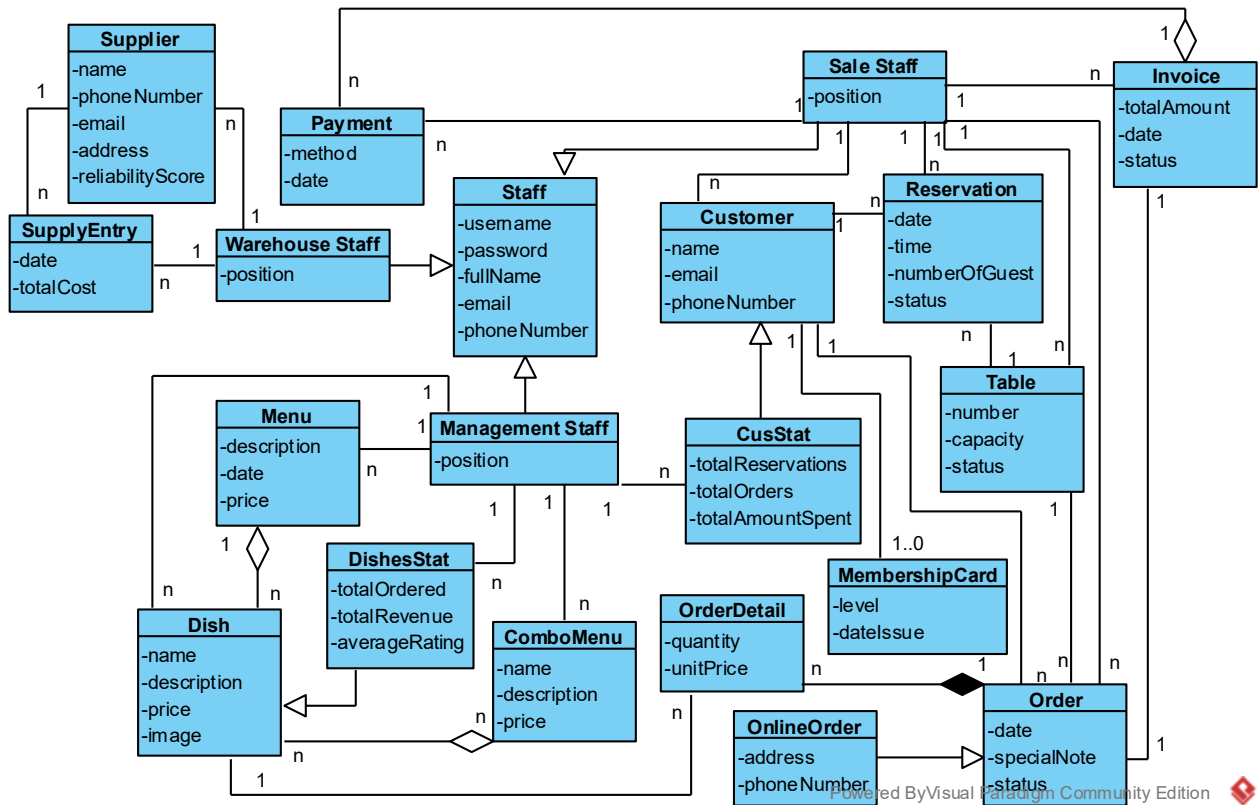
- Relationship: 1 - 1
- Type: Association (each dish has one statistics object).

4.18. Customer - CusStat

- Relationship: 1 - 1
- Type: Association (each customer has one statistics object).

4.19. ManagementStaff - Menu / Dish / ComboMenu/Statistic

- Relationship: 1 - N (management side)
- Type: Association (Management staff manages menus, dishes, and combo menus).



C. Class diagram for the module

I. Module *Customer searches for dish information*

1. Enter the system -> The main interface appeared -> Need a class: MainInterfaceView which has at least:

An option to choose to search for dish information -> subSearchForDish

2. Choose the option to search for dish -> The search interface appeared -> need a class : SearchDishView which has at least:

An input text to enter the keyword -> inKey

A search button -> subSearch

A result table which could be clicked in to choose a dish -> outsubListDish

3. Enter keyword and click search -> the system has to search all dishes whose name contains the entered keyword -> need a method:

Name: searchDish()

Input: a keyword

Output: a list of dish

Assign to class Dish

The result list will be displayed in the SearchDishView

4. Click on a dish to view detail information -> The dish information interface appeared -> need a class DishInformationView:

A table shows detail information about the dish -> outTableDishInformation

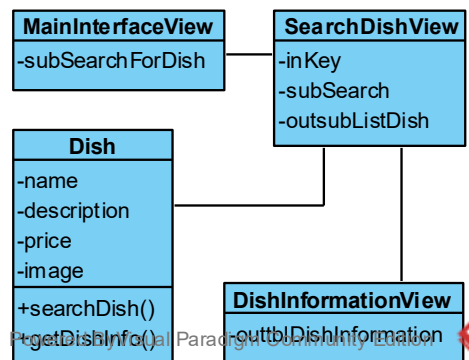
5. To get dish information -> the system has to collect data from the Dish class -> need a method:

Name: getDishInfo()

Input: dish ID

Output: a dish entity

Assign to class Dish



II. Module *Sale staff receives payment*

Enter the system -> The main interface appeared -> Need a class: **MainInterfaceView** which has at least:

A option to choose to login -> `subLoginInterface`

2. Choose the option to login -> The login interface appeared -> need a class **LoginView**:

Input for username -> `inUsername`

Input for password -> `inpassword`

A submit button -> `subLogin`

3. Enter the username and password -> the system must validate if the login is correct -> need a method:

Name: `checkLogin()`

Input: username, password

Output: Boolean

Assign to class: **Staff**

4. Once login is successfully -> the main staff interface is appeared -> need a class : **StaffMainInterfaceView** has at least:

An option to choose to process payment -> `subPayment`

5. Choose the option to search for table -> The search interface appeared -> need a class : **SearchTableView**:

An input text to enter the keyword -> `inKey`

A search button -> `subSearch`

A result table which could be clicked in to choose a table -> outsubListTable

6. Enter keyword and click search -> the system has to search all tables whose name contains the entered keyword -> need a method:

Name: searchTable()

Input: a keyword

Output: a list of table

Assign to class: Table

The result list will be displayed in the SearchTableView

7. Click on correct table to process payment

8. After clicking on table, the system displays the information including ordered dishes, unit price, subtotal, tax and total amount -> need a class: PaymentInvoicePreview has:

A table shows detail information about the temporary invoice -> outDetailTemporaryInvoice

An confirm button for Staff to confirm the invoice -> subConfirmInvoice

9. To get the order information for temporary invoice, the system needs a method:

Name: getOrderDetail()

Input: table id

Output: an Order entity

Assign to class: Order

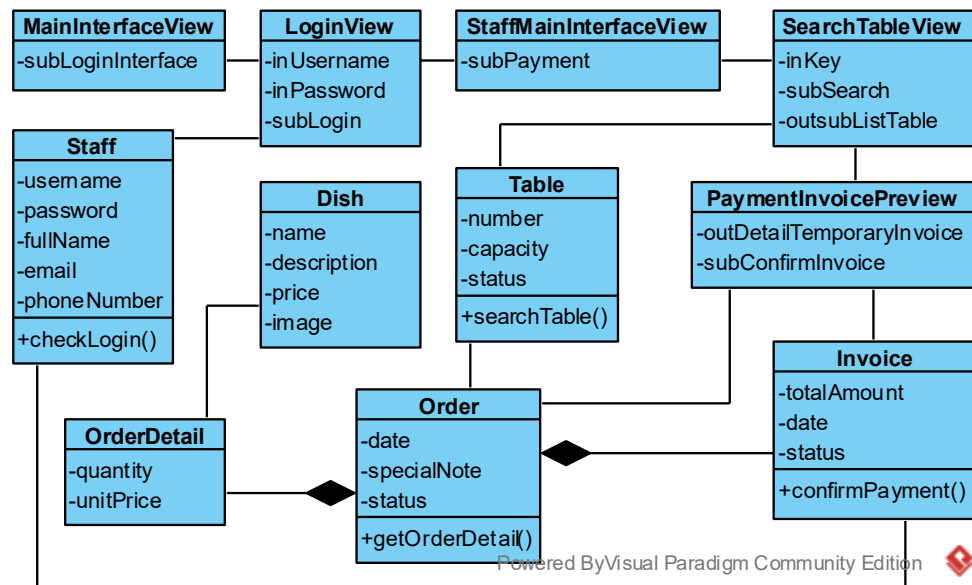
10. Staff reviews and confirms the temporary invoice -> Staff click on confirm button -> The system records the transaction, save invoice to Class Invoice, prints the invoice, updates the table status to “Available” -> need a method:

Name: confirmPayment()

Input: Order id

Output: a message to notify success or not

Assign to class: Invoice



D. State Diagram

I. Module *Customer searches for dish information*

Initial State:

- **Initial** -> MainInterfaceView

Transition 1:

- **From State:** MainInterfaceView
- **Event :** click SearchButton
- **To State:** SearchDishView

Transition 2:

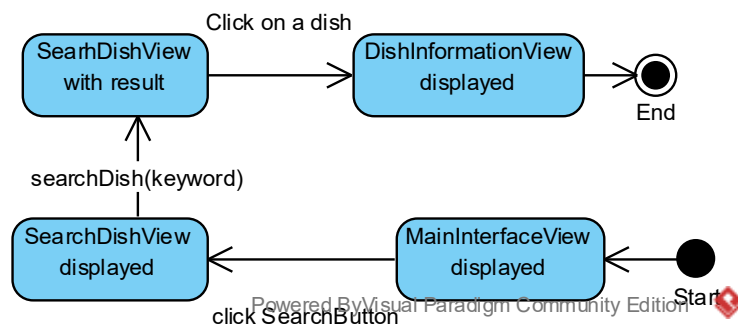
- **From State:** SearchDishView
- **Event :** searchDish(keyword)
- **Action:** results = searchDish(keyword); displayResults(results)
- **To State:** SearchDishView

Transition 3:

- **From State:** SearchDishView
- **Event :** select Dish
- **Action:** details = getDishInfo(dishID)
- **To State:** DishInformationView

Final Transitions:

- **From State:** DishInformationView
- **Event :** Close information window or click back
- **To State:** End



II. Module *Sale staff receives payment*

Initial State:

- **Initial** -> MainInterfaceView

Transition: Access Login

- **Current State:** MainInterfaceView
- **Event:** click subLoginInterface()
- **To State:** LoginView

Transition: Perform Login

- **Current State:** LoginView
- **Event:** click subLogin(username, password)
- **Action:** checkLogin()
- **To State:** StaffMainInterface

Transition: Access Payment Module

- **Current State:** StaffMainInterface
- **Event:** click subPayment()
- **To State:** SearchTableView

Transition: Search Table

- **Current State:** SearchTableView
- **Event:** click subSearch(keyword)
- **Action:** list = searchTables(keyword); displayResults(list)
- **To State:** SearchTableView

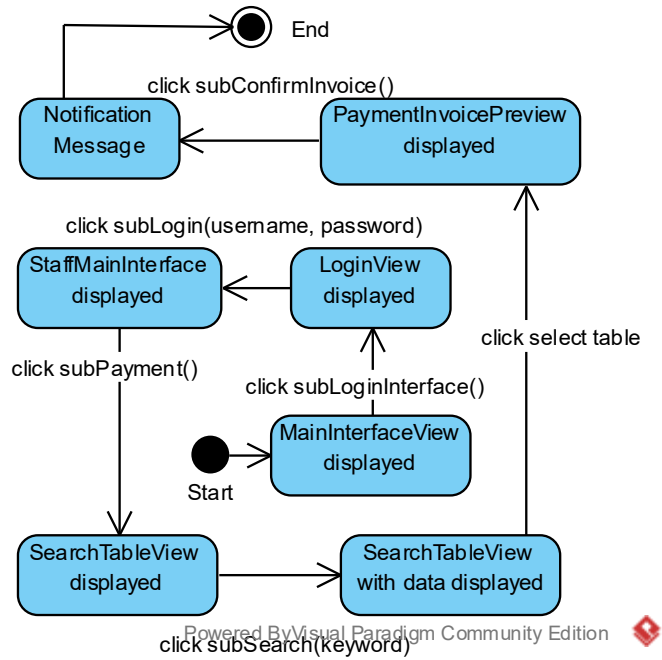
Transition: Select Table

- **Current State:** SearchTableView
- **Event:** click select table
- **Action:** preview invoice = getOrderDetail (tableID)
- **To State:** PaymentInvoicePreview

7. Transition: Confirm Payment

- **Current State:** PaymentInvoicePreview

- **Event :** click subConfirmInvoice()
- **Action:** processPayment(); updateInvoiceStatus(); printInvoice()
- **To State:** PaymentSuccessNotification

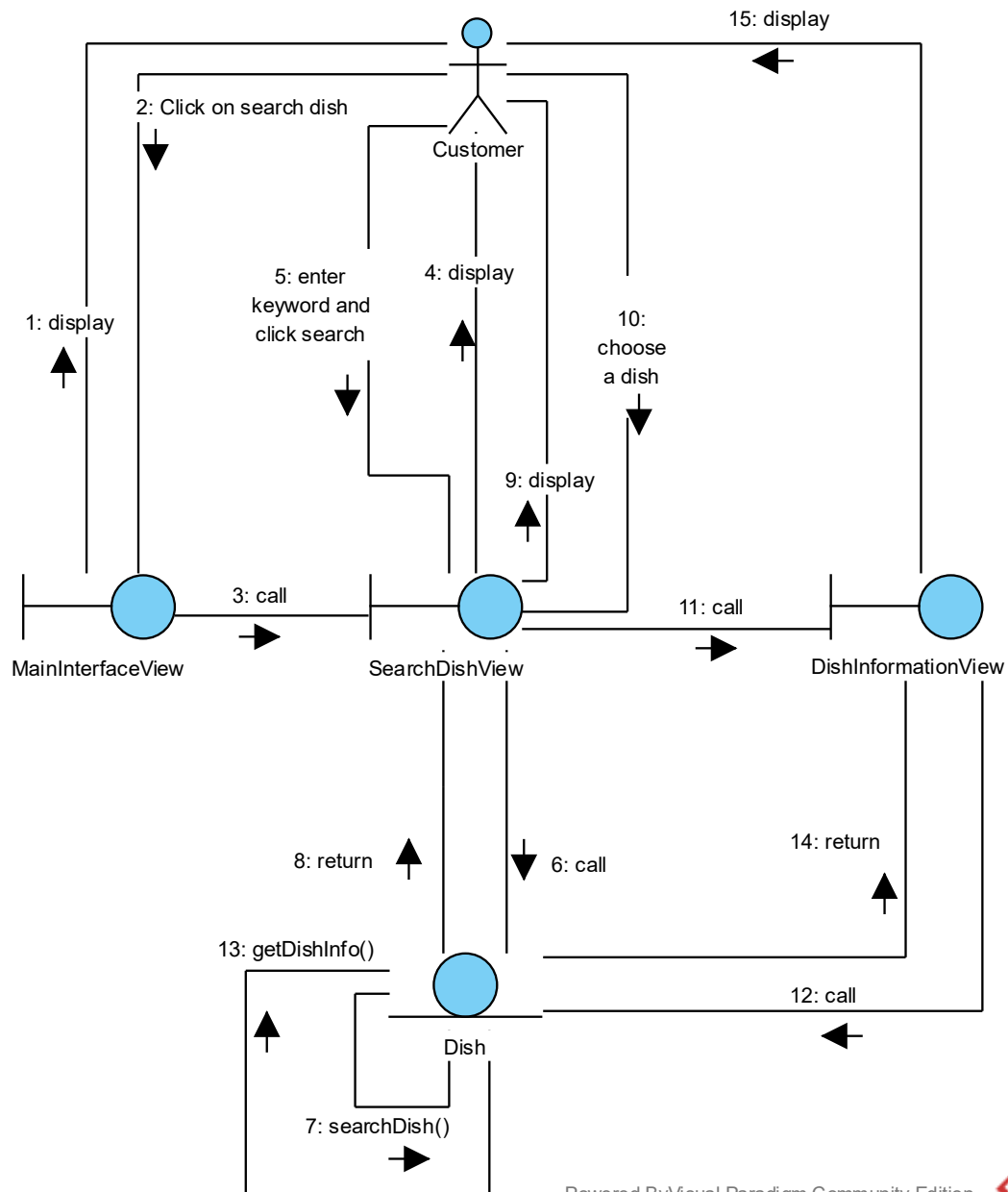


E. Scenarios version 2

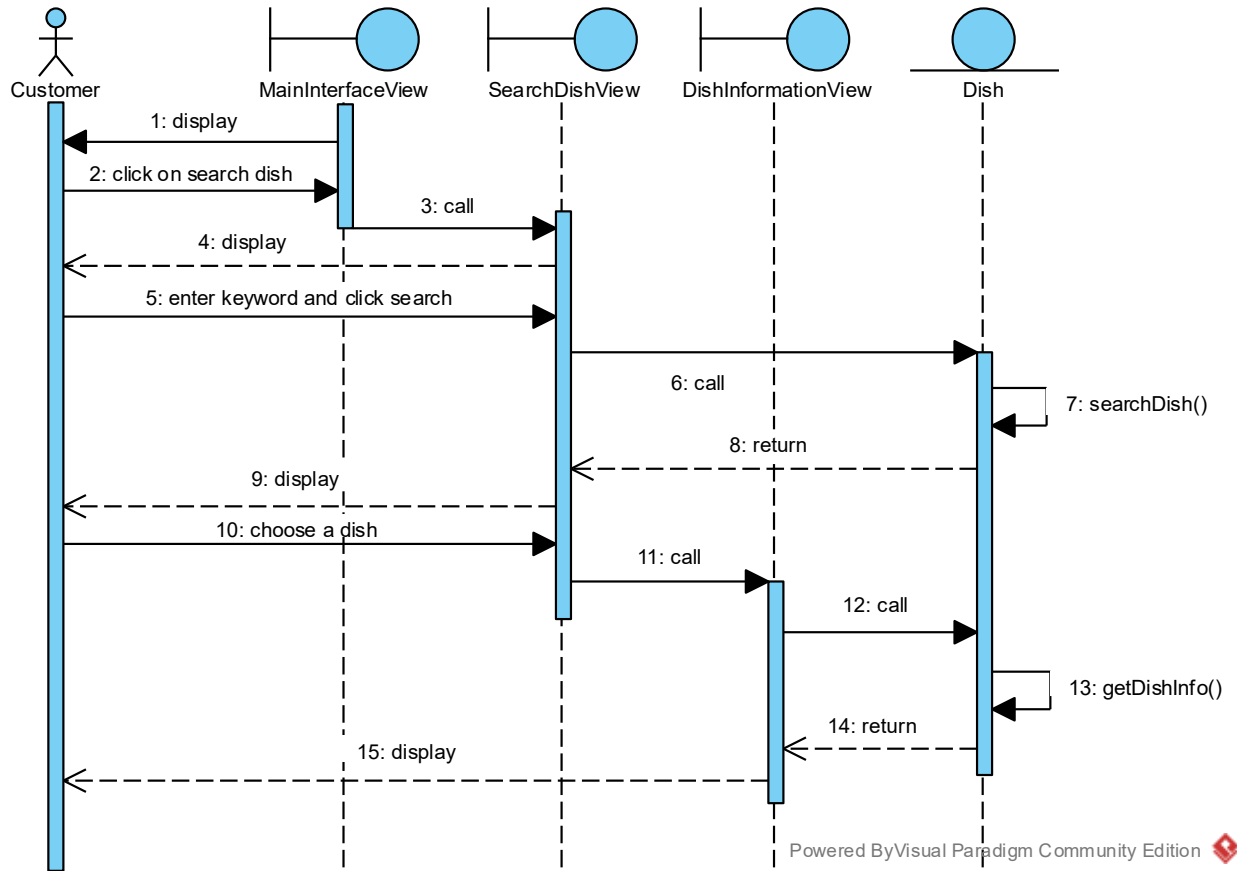
I. Module *Customer searches for dish information*

1. Enter the system, the main interface displays itself to the user
2. From the main interface, the user selects the “Search dish” function
3. The class MainInterfaceView calls the class SearchDiskView
4. The class SearchDiskView displays itself to the user
5. The user enters keyword and click on the search button
6. The class SearchDiskView calls the class Dish to process
7. The class Dish calls the method searchDish()
8. The class Dish returns the results to the class SearchDiskView
9. The class SearchDiskView displays the results to the user
10. The user choose a dish to view detail information
11. The class SearchDiskView calls the class DishInformationView
12. The class DishInformationView calls the class Dish to process
13. The class Dish calls the method getDishInfo()
14. The class Dish returns the results to the class DishInformationView
15. The class DishInformationView displays the results to the user

Communication diagram



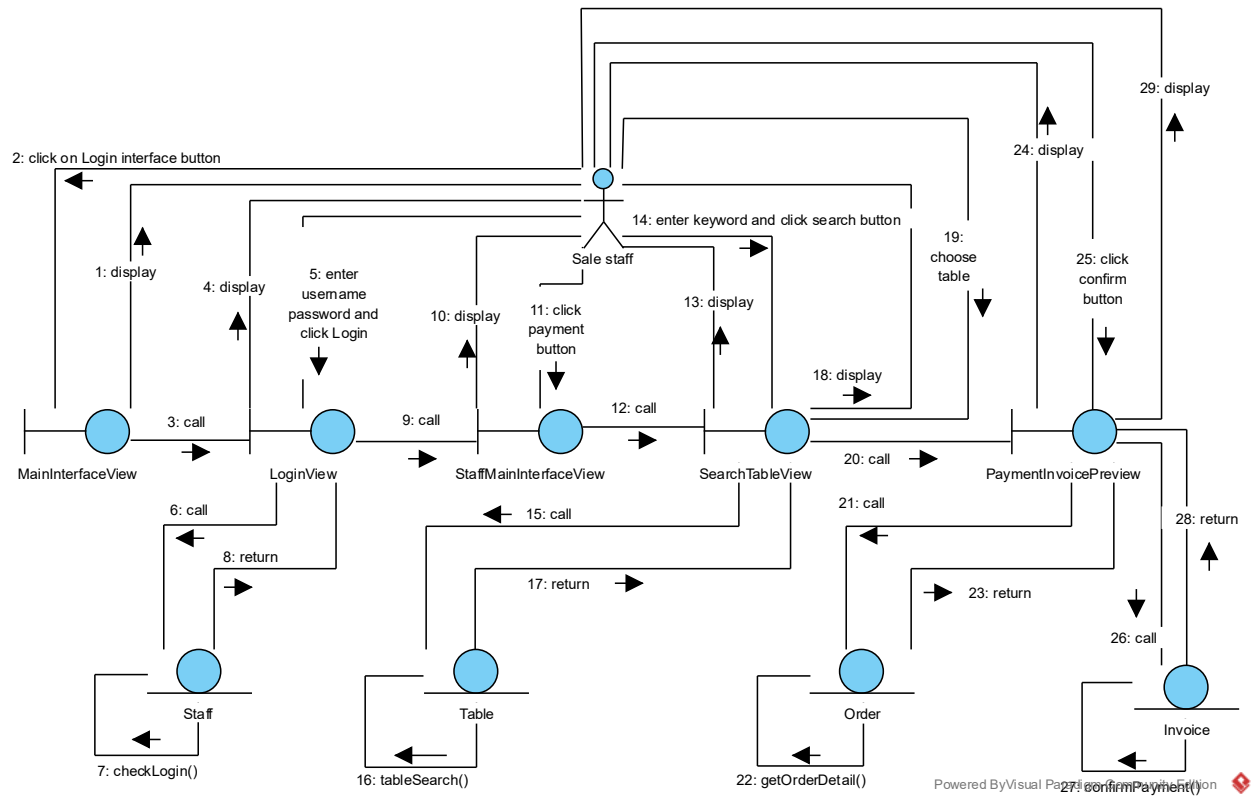
Sequence diagram



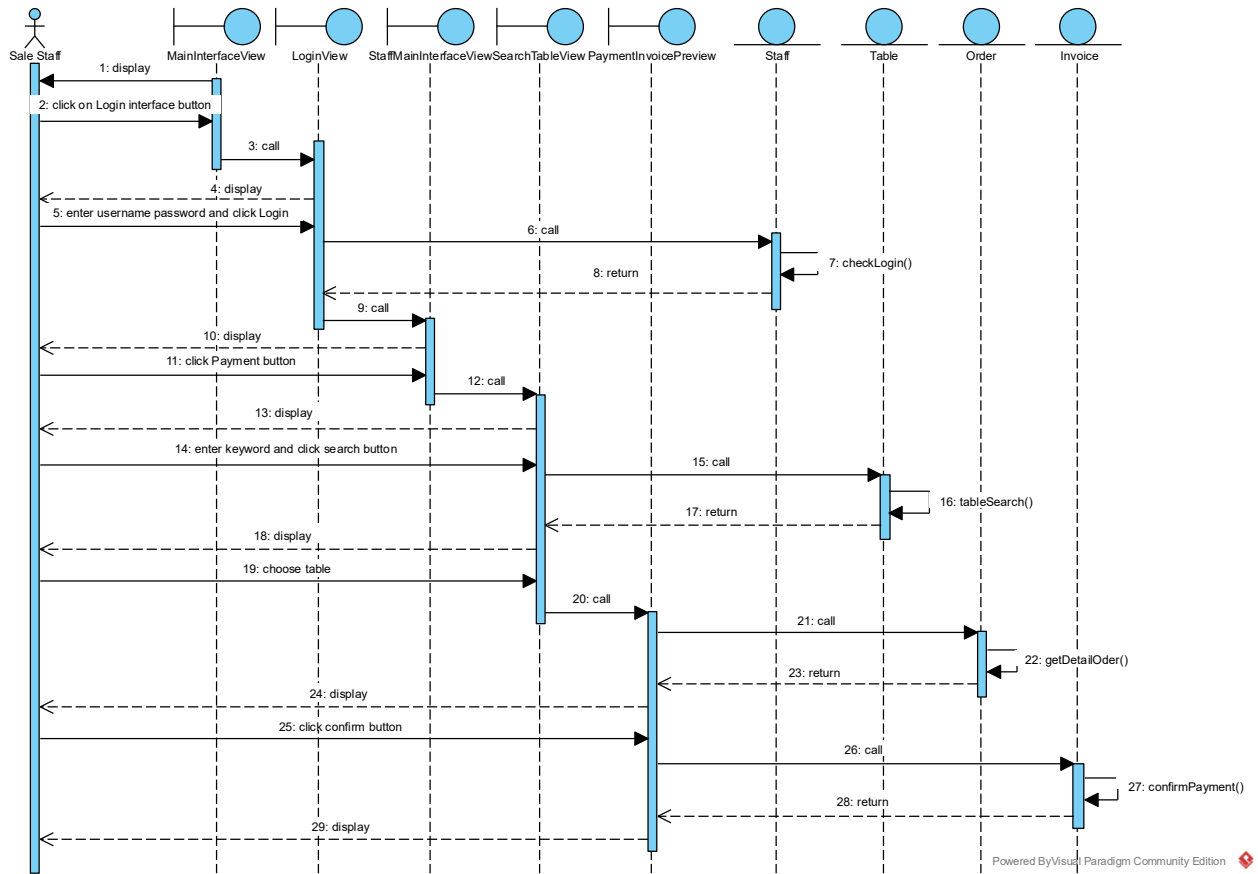
II. Module *Sale staff receives payment*

1. Enter the system, the main interface displays itself to the sale staff
2. From the main interface, the sale staff selects the Login interface button
3. The class `MainInterfaceView` calls the class `LoginView`
4. The class `LoginView` displays itself to the sale staff
5. The sale staff enters username and password and then clicks on the Login Button
6. The class `LoginView` calls the class `Staff` to process
7. The class `Staff` calls the method `checkLogin()`, the login successful
8. The class `Staff` returns the result to class `LoginView`
9. The class `LoginView` calls the class `StaffMainInterfaceView`
10. The class `StaffMainInterfaceView` displays itself to the sale staff
11. The sale staff clicks on the payment button
12. The class `StaffMainInterfaceView` calls the class `SearchTableView`
13. The class `SearchTableView` displays itself to the sale staff
14. The sale staff enters keyword and click on search button
15. The class `SearchTableView` calls the class `Table` to process
16. The class `Table` calls the method `searchTable()`
17. The class `Table` returns the result to the class `SearchTableView`
18. The class `SearchTableView` displays the result to the sale staff
19. The sale staff chooses a table to process payment
20. The class `SearchTableView` calls the class `PaymentInvoicePreview`
21. The class `PaymentInvoicePreview` calls the class `Order` to process
22. The class `Order` calls the method `getOrderDetail()`
23. The class `Order` returns the result to the class `PaymentInvoicePreview`
24. The class `PaymentInvoicePreview` displays itself to the sale staff
25. The sale staff clicks on confirm button to confirm the invoice
26. The class `PaymentInvoicePreview` calls the class `Invoice` to process
27. The class `Invoice` calls the method `confirmPayment()`
28. The class `Invoice` return result to the class `PaymentInvoiceView`
29. The class `PaymentInvoiceView` displays itself to the sale staff

Communication diagram



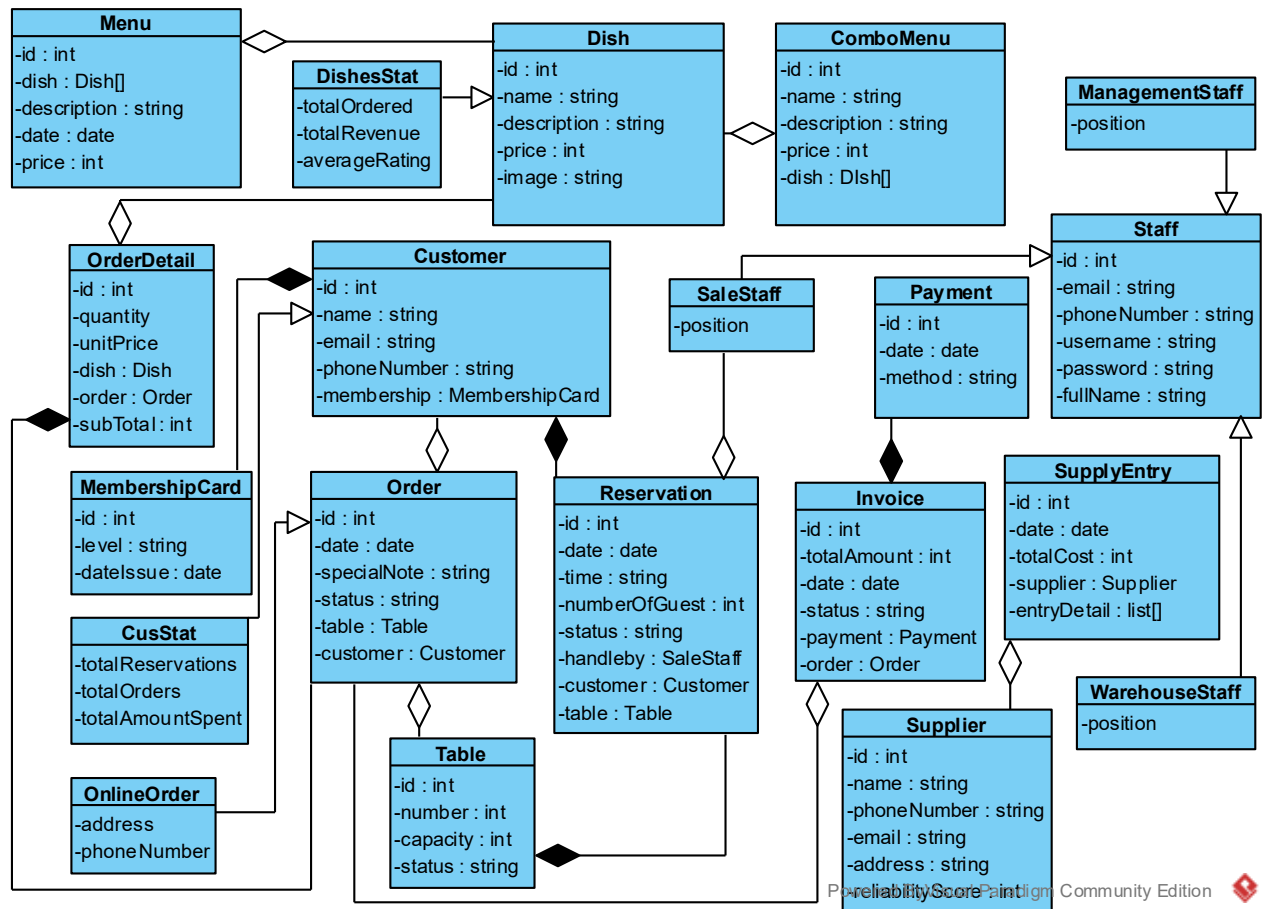
Sequence diagram



DESIGN

A. Entity Class Diagram

1. Add id to all classes, except: Management Staff, Warehouse Staff, Sale Staff, CusStat, Online Order, DishStat
2. Add data type to attribute
3. There is no relationship need conversion



B. Database Design

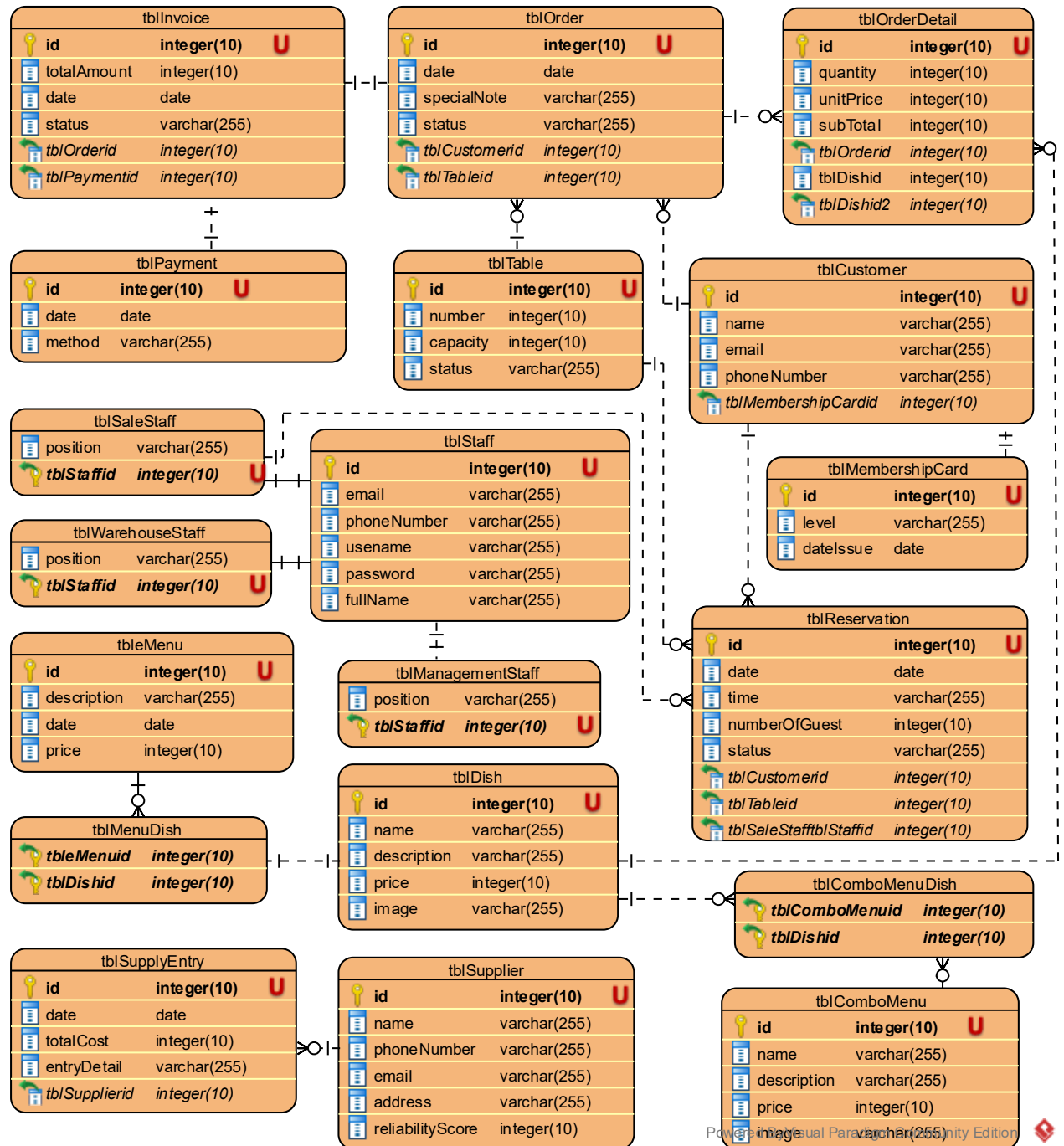
1. Convert entity class to ERD

- Class Staff -> tblStaff: id, email, phoneNumber, username, password, fullName
- Class ManagementStaff -> tblManagementStaff: position
- Class SaleStaff -> tblSaleStaff: position
- Class WarehouseStaff -> tblWarehouseStaff: position
- Class ComboMenu -> tblComboMenu: id, name
- Class Dish -> tblDish: id, name, description, price, image
- Class Menu -> tblMenu: id, dish, description, date, price
- Class Payment -> tblPayment: id, date, method
- Class Invoice -> tblInvoice: id, totalAmount, date, status
- Class Supplier -> tblSupplier: id, name, phoneNumber, email, address, reliabilityScore
- Class SupplyEntry -> tblSupplyEntry: id, date, totalCost, entryDetail
- Class Customer -> tblCustomer: id, name, name, email, phoneNumber
- Class OrderDetail -> tblOrderDetail: id, quantity, unitPrice
- Class Reservation -> tblReservation: id, date, time, numberOfGuest, status
- Class Table -> tblTable: id, number, capacity, status
- Class Order -> tblOrder: id, date, specialNote, status
- Class MembershipCard -> tblMembershipCard: id, level, dateIssue

2. Relationship cardinality mapping

- 1 tblStaff – 1 tblSaleStaff -> tblSaleStaff has foreign key tblStaffId
- 1 tblStaff – 1 tblWarehouseStaff -> tblWarehouseStaff has foreign key tblStaffId
- 1 tblStaff – 1 tblManagementStaff -> tblManagementStaff has foreign key tblStaffId
- n tblMenu – n tblDish -> Create tblMenuDish
- n tblComboMenu – n tblDish -> Create tblComboMenuDish
- 1 tblCustomer – n tblOrder -> tblOrder has foreign key tblCustomerId
- 1 tblCustomer – n tblReservation -> tblReservation has foreign key tblCustomerId
- 1 tblCustomer – 1 tblMembershipCard -> tblMembershipCard has foreign key tblCustomerId
- 1 tblOrder – n tblOrderDetail -> tblOrderDetail has foreign key tblOrderId
- 1 tblDish – n tblOrderDetail -> tblOrderDetail has foreign key tblDishId
- 1 tblOrder – 1 tblInvoice -> tblInvoice has foreign key tblOrderId
- 1 tblInvoice – 1 tblPayment -> tblPayment has foreign key tblInvoiceId
- 1 tblSaleStaff – n tblInvoice -> tblInvoice has foreign key tblSaleStaffId
- 1 tblSupplier – n tblSupplyEntry -> tblSupplyEntry has foreign key tblSupplierId

- 1 tblTable – n tblOrder -> tblOrder has foreign key tblTableId
- 1 tblTable – n tblReservation -> tblReservation has foreign key tblTableId
- n tblReservation – 1 tblTable -> tblReservation has foreign key tblTableId



C. Class Diagram for Module

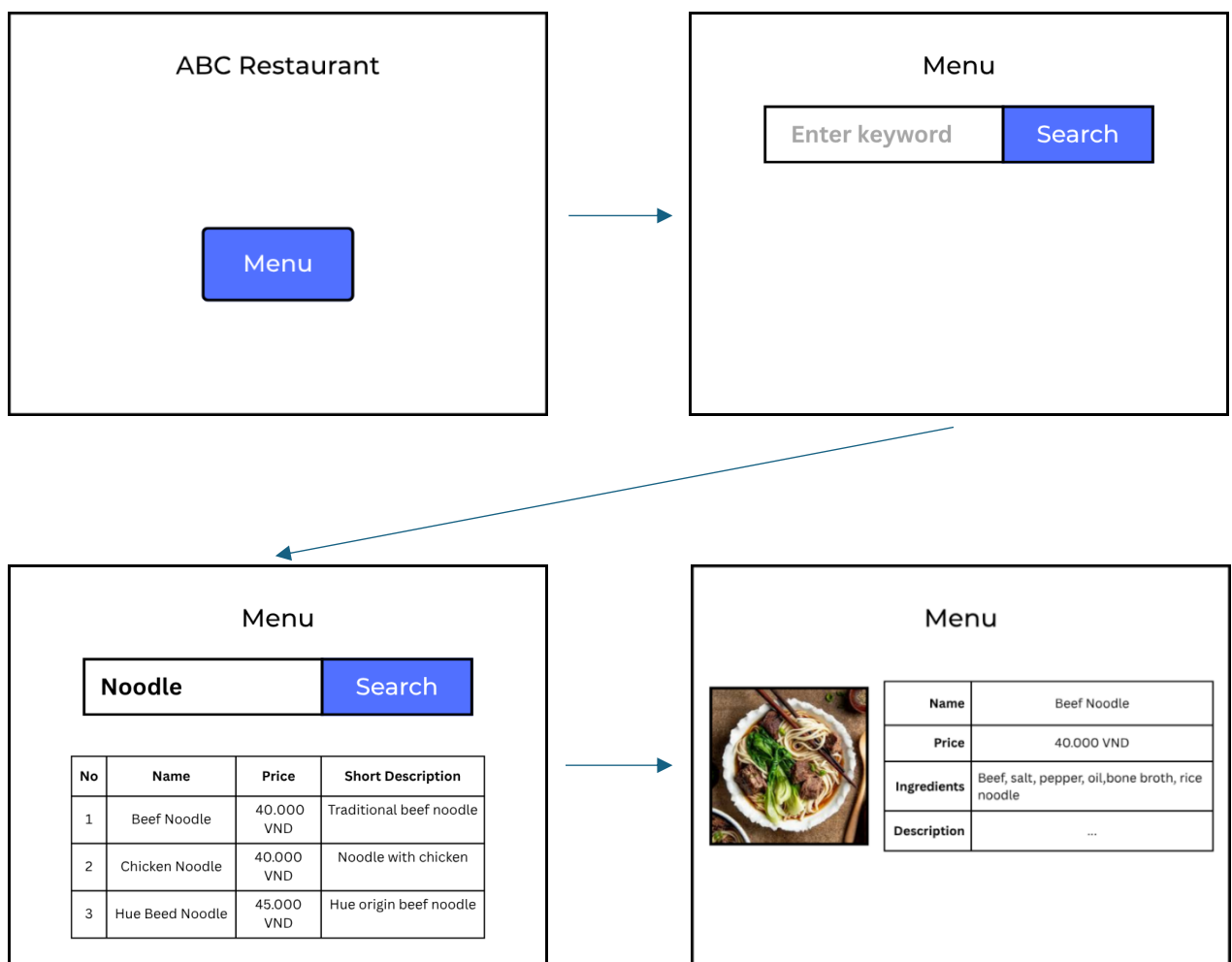
1. Customer searches for dish information

1.1. Interface

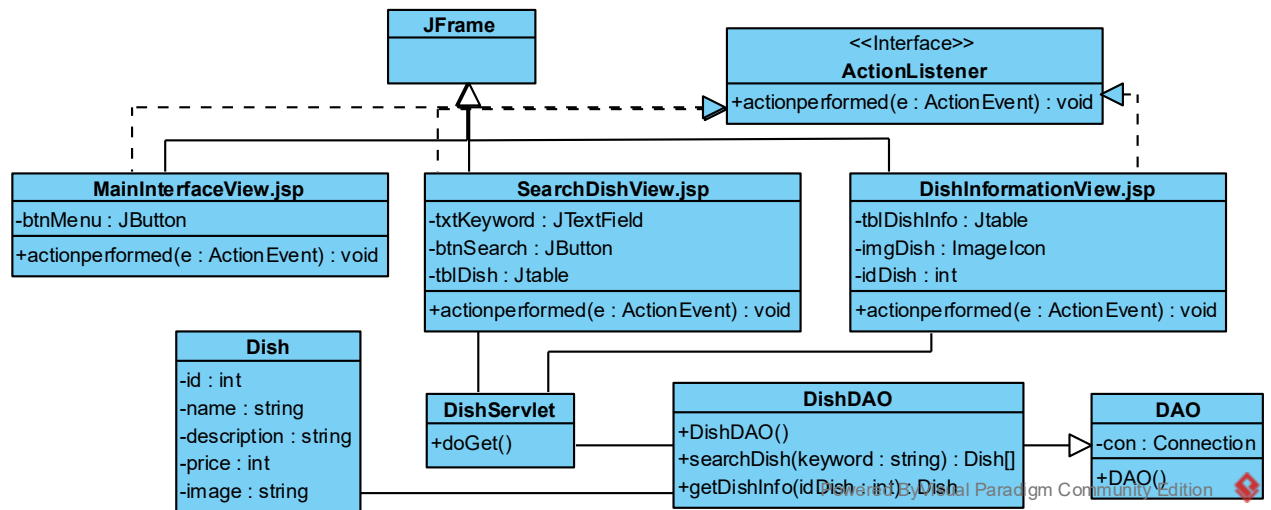
View Interface Layer includes: MainInterfaceView.jsp, SearchDishView.jsp, DishInformationView.jsp

Data Control Layer includes: DishDAO

Controller Layer includes: DishServlet



1.2. Activity Diagram



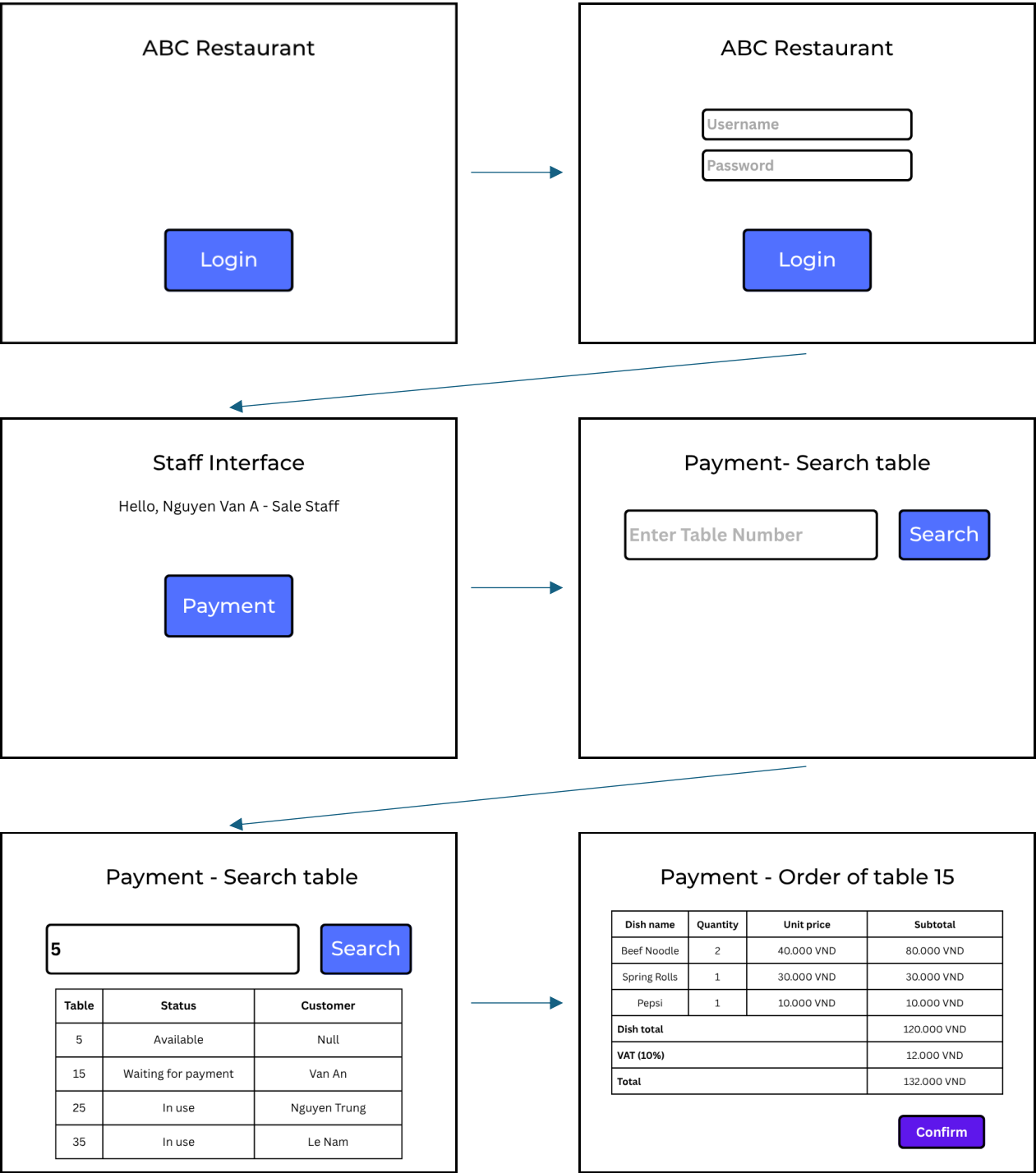
2. Sale staff receives payment

2.1. Interface

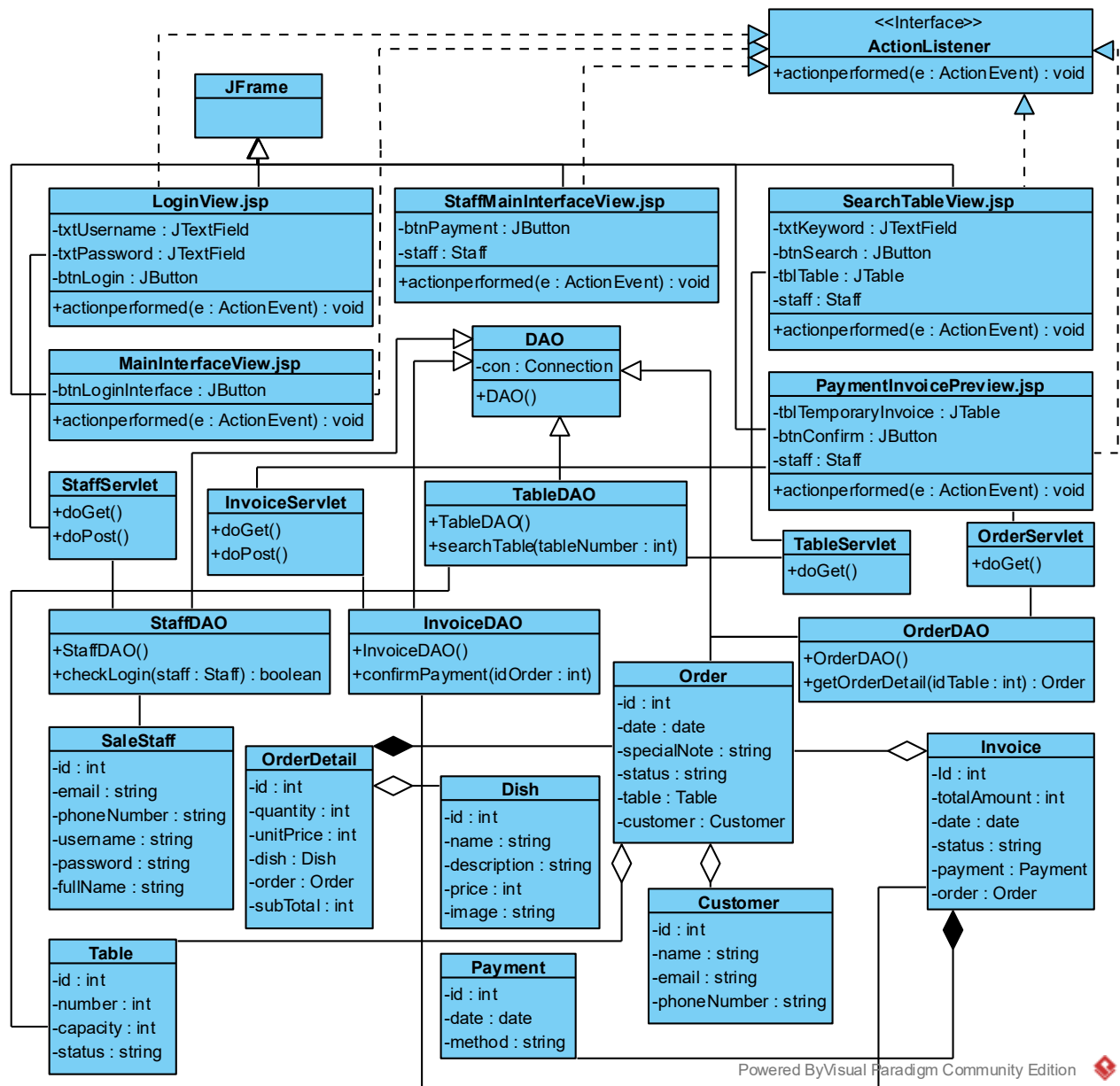
View Interface Layer include: MainInterfaceView.jsp, LoginView.jsp, StaffMainInterfaceView.jsp, SearchTableView.jsp, PaymentInvoicePreview.jsp

Data Control Layer: StaffDAO, InvoiceDAO, TableDAO, OrderDAO

Controller Layer includes: TableServlet, OrderServlet, StaffServlet, InvoiceServlet

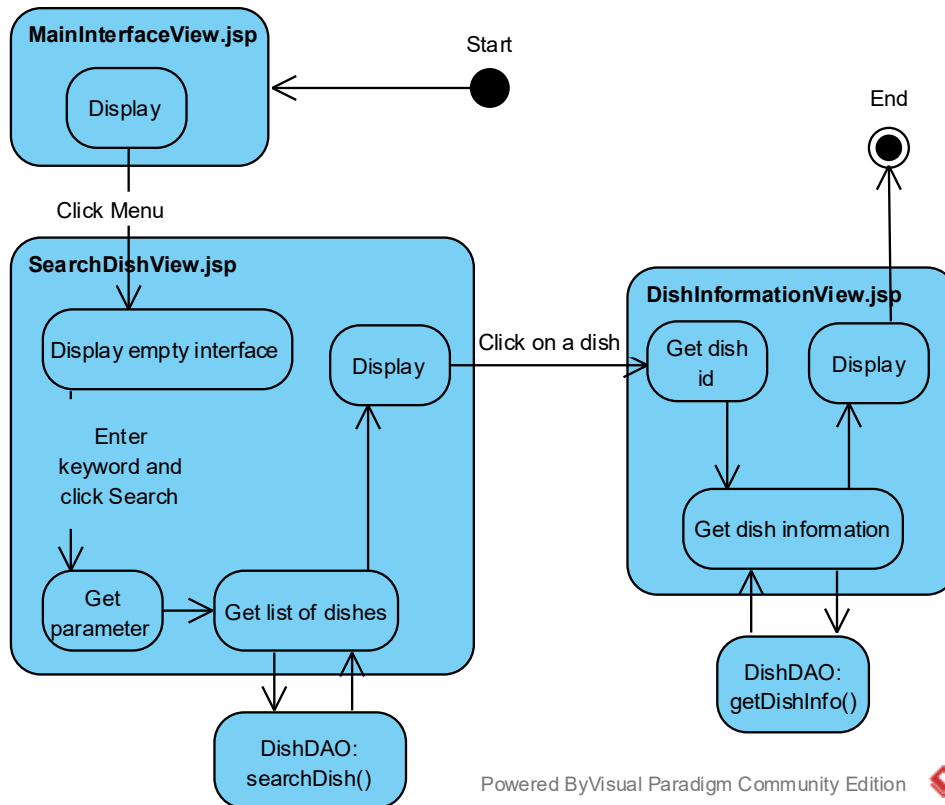


2.2. Activity Diagram

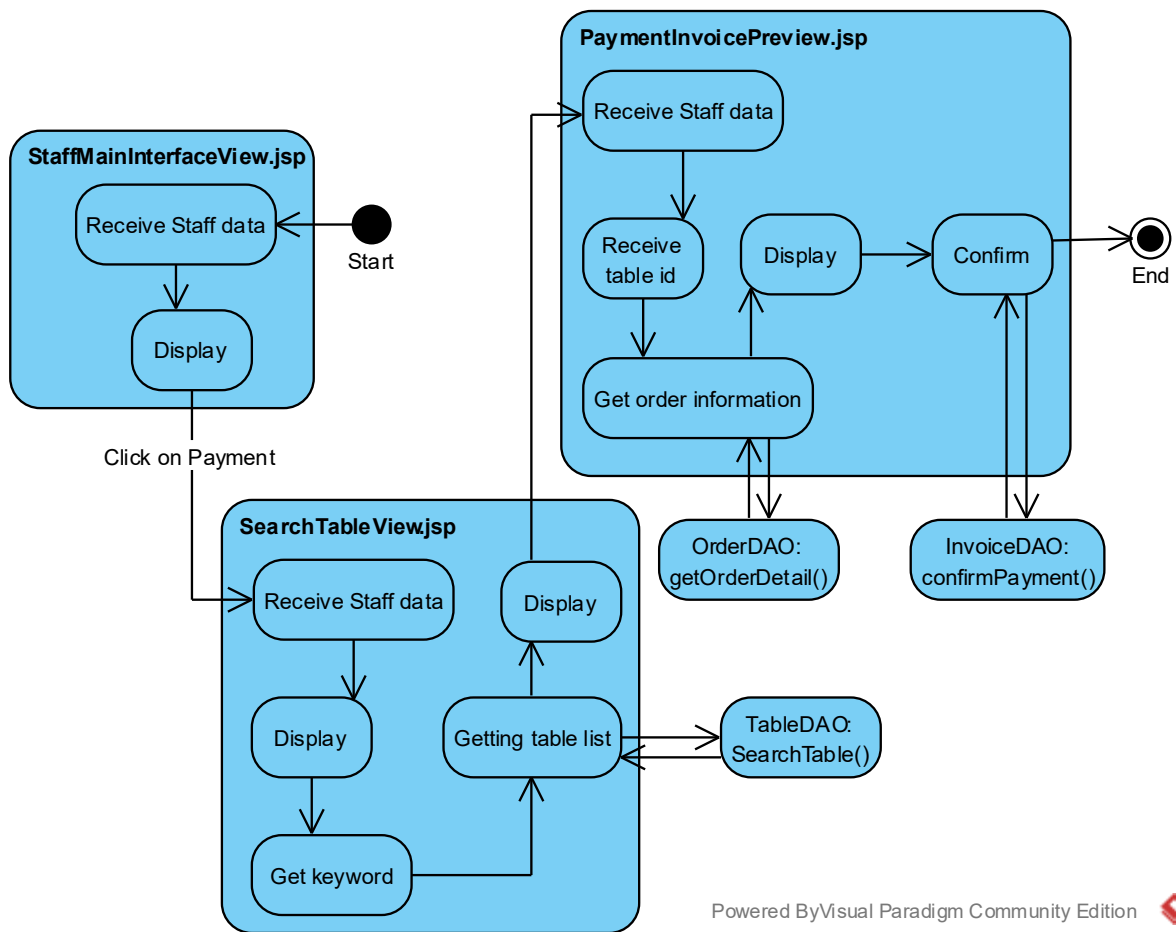


D. Active Diagram for Module

1. Customer searches for dish information



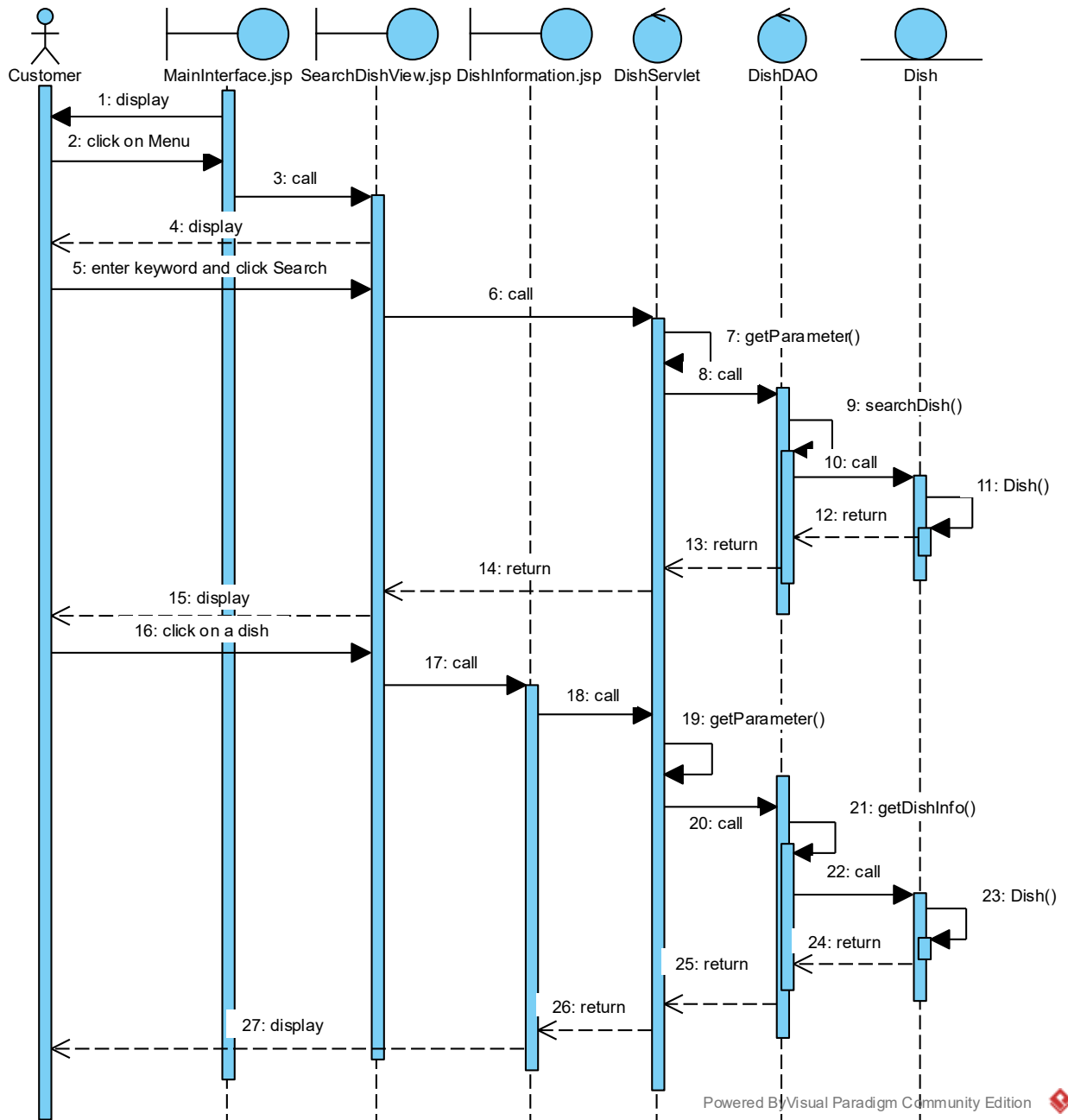
2. Sale staff receives payment



E. Scenario v3.0

1. Customer searches for dish information

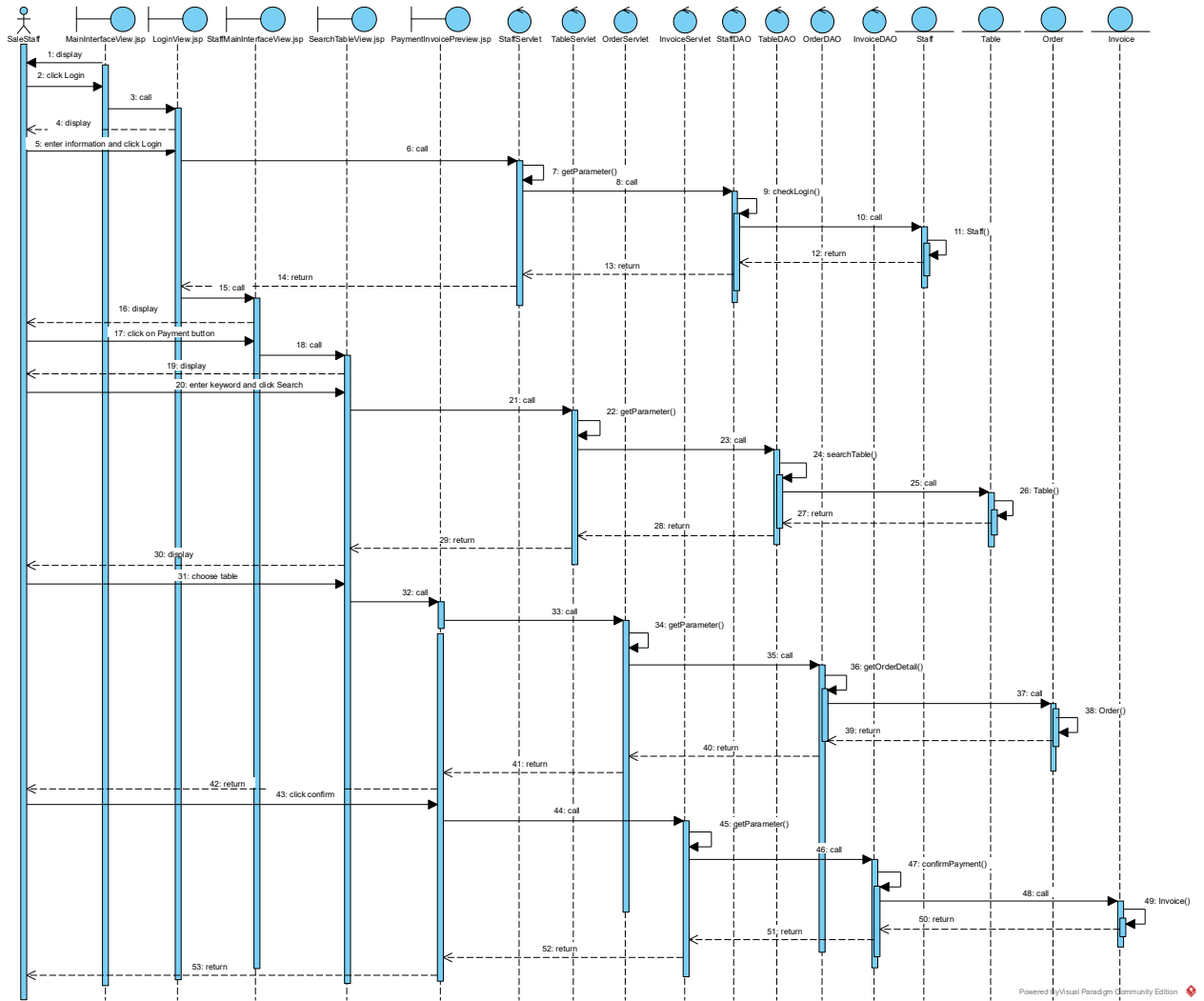
1. User enters the system, MainInterfaceView.jsp displays itself to the user
2. In MainInterfaceView.jsp, the user clicks on the Menu button
3. MainInterfaceView.jsp calls SearchDishView.jsp
4. SearchDishView.jsp displays itself to the user
5. The user enters a keyword and clicks the Search button
6. SearchDishView.jsp sends request to DishServlet
7. DishServlet get parameter
8. DishServlet calls the class DishDAO to process the request
9. The class DishDAO calls the method searchDish()
10. The method searchDish() executes and calls the class Dish to create data objects
11. The class Dish packs the dish data into objects
12. The class Dish returns the objects to the method searchDish()
13. The method searchDish() returns the result to DishServlet
14. DishServlet forward to SearchDishView.jsp
15. SearchDishView.jsp displays the list of dishes to the user
16. The user clicks on a specific dish
17. SearchDishView.jsp calls DishInformationView.jsp
18. DishInformationView.jsp sends request to DishServlet
19. DishServlet gets parameter
20. DishServlet calls the class DishDAO to process the request
21. The class DishDAO calls the method getDishInfo()
22. The method getDishInfo() executes and calls the class Dish to create data objects
23. The class Dish packs the dish information data
24. The class Dish returns the object to the method getDishInfo()
25. The method getDishInfo() returns the result to DishServlet
26. DishServlet forward to DishInformation.jsp
27. DishInformationView.jsp displays the dish details to the user



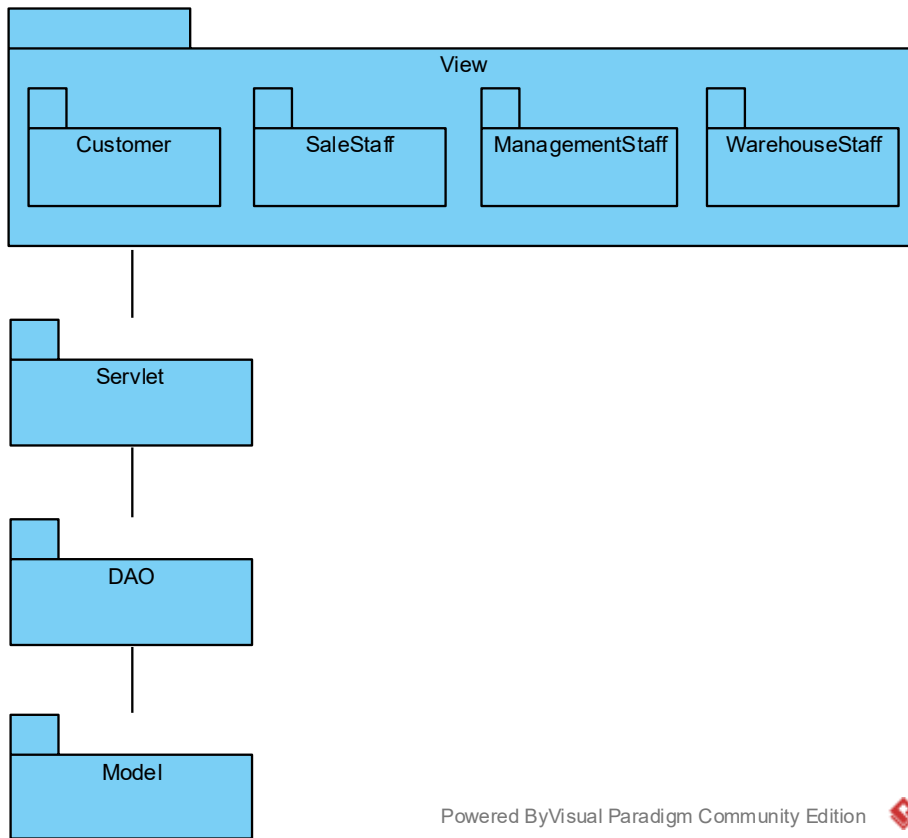
2. Sale staff receives payment

1. Enter the system, MainInterfaceView.jsp displays itself to the sale staff
2. From the main interface, the sale staff selects the Login interface button
3. MainInterfaceView.jsp calls LoginView.jsp
4. LoginView.jsp displays itself to the sale staff
5. The sale staff enters username and password and then clicks on the Login Button
6. LoginView.jsp sends request to StaffServlet
7. StaffServlet gets parameter
8. StaffServlet calls the class StaffDAO to process the request
9. The class StaffDAO calls the method checkLogin()
10. The method checkLogin() call Staff to validates the login information, If the information is valid, the class Staff create data object
11. The class Staff packs data into staff objects
12. The class Staff returns the objects to the method checkLogin()
13. The method checkLogin() returns the result to StaffServlet
14. StaffServlet forward result to LoginView.jsp
15. LoginView.jsp calls StaffMainInterfaceView.jsp
16. StaffMainInterfaceView.jsp displays itself to the sale staff
17. The sale staff clicks on the Payment button
18. StaffMainInterfaceView.jsp calls SearchTableView.jsp
19. SearchTableView.jsp displays itself to the sale staff
20. The sale staff enters keyword and click on Search button
21. SearchTableView.jsp sends request to TableServlet
22. TableServlet get parameter
23. TableServlet calls the class TableDAO to process the request
24. The class TableDAO calls the method searchTable()
25. The method searchTable() executes and calls the class Table to create data objects
26. The class Table packs the table data into objects
27. The class Table returns the result to the method searchTable()
28. The method searchTable() returns the result to TableServlet
29. TableServlet forward result to SearchTableView.jsp
30. SearchTableView.jsp displays the result to sale staff
31. The sale staff chooses a table to process payment
32. SearchTableView.jsp calls PaymentInvoicePreview.jsp
33. PaymentInvoicePreview.jsp sends request to OrderServlet
34. OrderServlet get parameter
35. OrderServlet calls the class OrderDAO to process the request
36. The class OrderDAO calls the method getOrderDetail()
37. The method getOrderDetail() executes and calls the class Order to create data objects

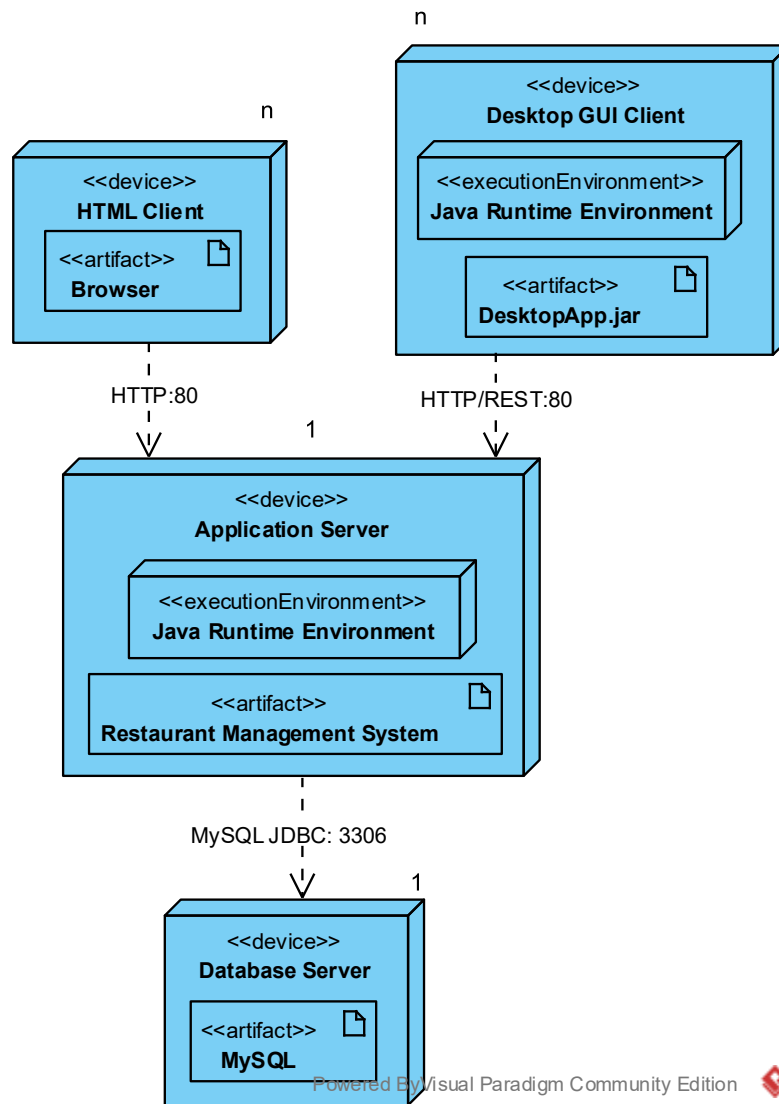
38. The class Order packs the order data into object
39. The class Order returns the result to the method getOrderDetail()
40. The method getOrderDetail() returns the result to OrderServlet
41. OrderServlet forward the result to PaymentInvoicePreview.jsp
42. PaymentInvoicePreview.jsp displays itself to the sale staff
43. The sale staff clicks on confirm button to confirm the invoice
44. PaymentInvoicePreview.jsp sends request to InvoiceServlet
45. InvoiceServlet get parameter
46. InvoiceServlet calls the class InvoiceDAO to process the request
47. The class InvoiceDAO calls the method confirmPayment()
48. The method confirmPayment() executes and call the class Invoice to create data object
49. The class Invoice packs the invoice data into object
50. The class Invoice return result object to method confirmPayment()
51. The method confirmPayment() save data to the database and return success message to InvoiceServlet
52. InvoiceServlet forward the message to PaymentInvoicePreview.jsp
53. PaymentInvoicePreview.jsp displays the message



F. Package Diagram



G. Deployment Diagram



H. Coding

MainInterface.jsp

Bet88 Restaurant Management System

Browse Menu

Explore our delicious dishes

Search Dishes

Staff Login

Access staff portal

Staff Login

Module 1: Customer searches for dish information

SearchDishView.jsp

Menu

← Back to Home

Find Your Favorite Dish

Enter dish name (e.g., Noodle, Soup, Rice)

Search

Menu

← Back to Home

Find Your Favorite Dish

Enter dish name (e.g., Noodle, Soup, Rice)

Search

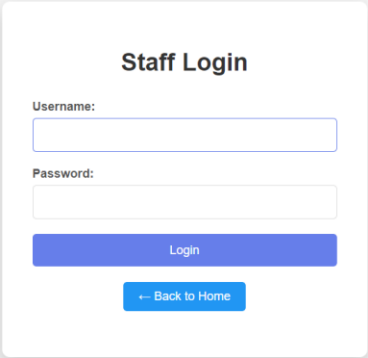
Dishes with keyword "rice" (1 dishes found)

No.	Image	Name	Price	Short Description
1		Fried Rice	40,000 VND	Delicious fried rice with eggs, vegetables (carrot...

DishInformation.jsp

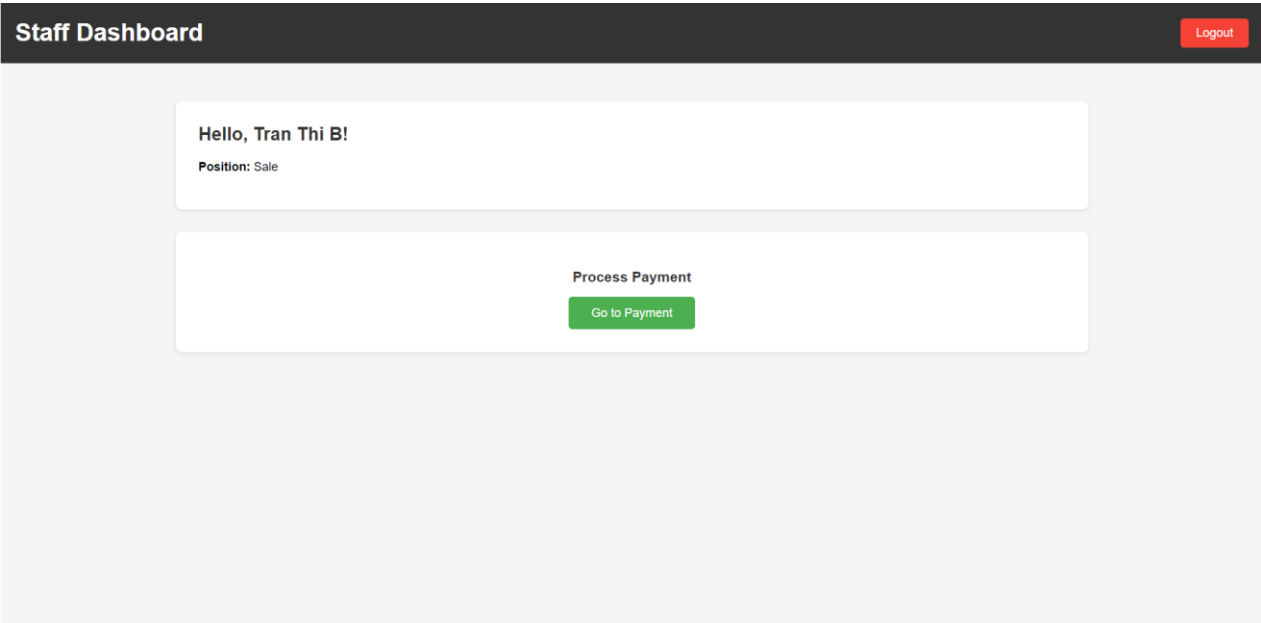
Module 2: Sale staff receives payment

LoginView.jsp



A login form titled "Staff Login" centered on a light gray background. The form is a white box with a shadow. It contains two input fields: "Username:" and "Password:". Below the password field is a blue "Login" button. At the bottom of the form is a smaller blue button with a left arrow and the text "Back to Home".

StaffMainInterface.jsp



A dashboard interface titled "Staff Dashboard" in a dark header bar. In the top right corner of the header is a red "Logout" button. The main content area is light gray and contains two white boxes. The first box displays a greeting "Hello, Tran Thi B!" and the user's "Position: Sale". The second box, below the first, is titled "Process Payment" and contains a green "Go to Payment" button.

Search Table for Payment

[← Back to Dashboard](#)

Find Table by Number

Enter table number

Search

Search Table for Payment

[← Back to Dashboard](#)

Find Table by Number

Enter table number

Search

Search Results (1 tables found)

No.	Table Number	Capacity	Status	Action
1	Table 1	4 seats	Occupied	View Order

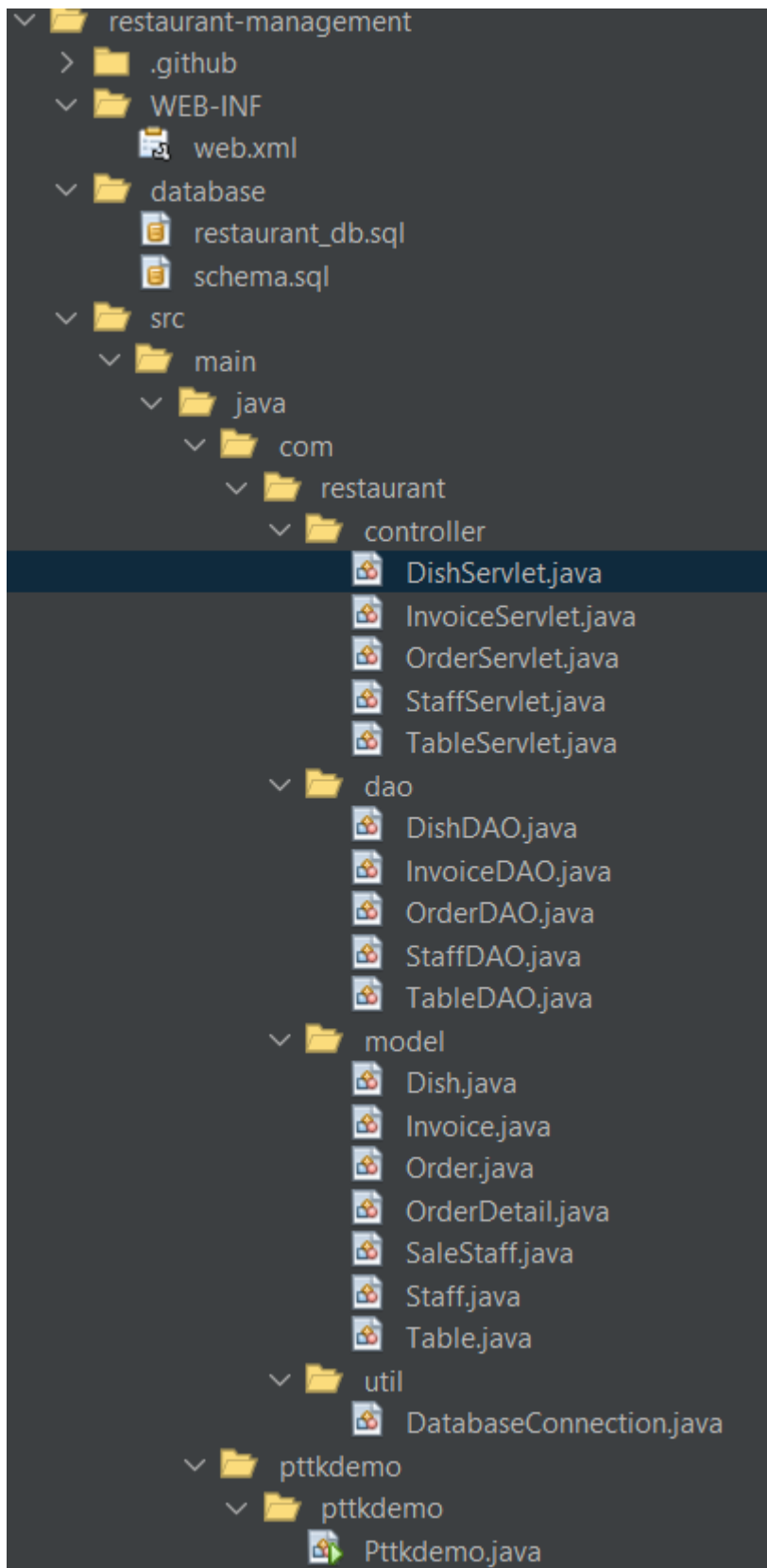
Payment Invoice Preview

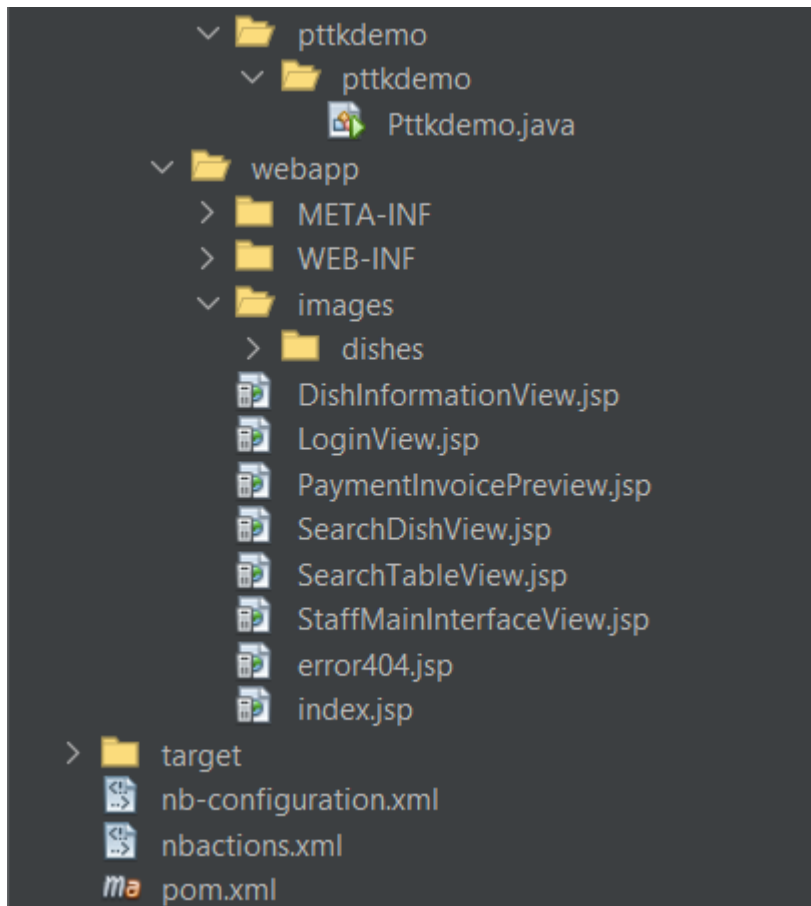
[← Back to Table Search](#)

✓ ✓ Payment confirmed successfully! Invoice has been created. Total: 82500 VND

[Process another payment](#)

Project file structure





Source Code:

<https://drive.google.com/file/d/1HgrbbFZsU7W7TdJTFvLfLWQfPuP6v09u/view?usp=sharing>

Spare link: <https://github.com/rustb0ne/pttkdemo>